Sensing and the Imaging Process (IV)

Introduction to Computational and Biological Vision

CS 202-1-5261

Computer Science Department, BGU

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What is an “image”?

How does an image form?

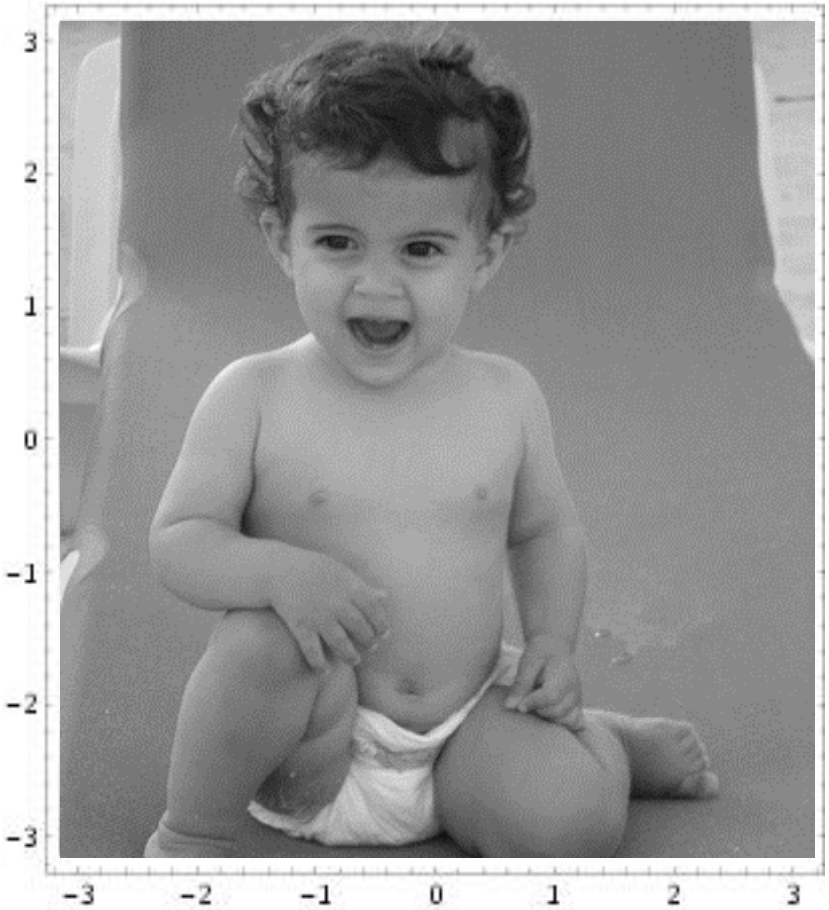
Image formation and representation

What is an image (Take 1)

$$I(x, y): R^2 \rightarrow R^3$$

I represents:

- Light Intensity/luminance
- Color*
- X ray radiation
- Temperature
- ⋮
- Any (often scalar*) measure of interest



∗ Sometimes, a vector quantity

Image formation and representation

Aspects of image formation

1. Geometry
2. Optics

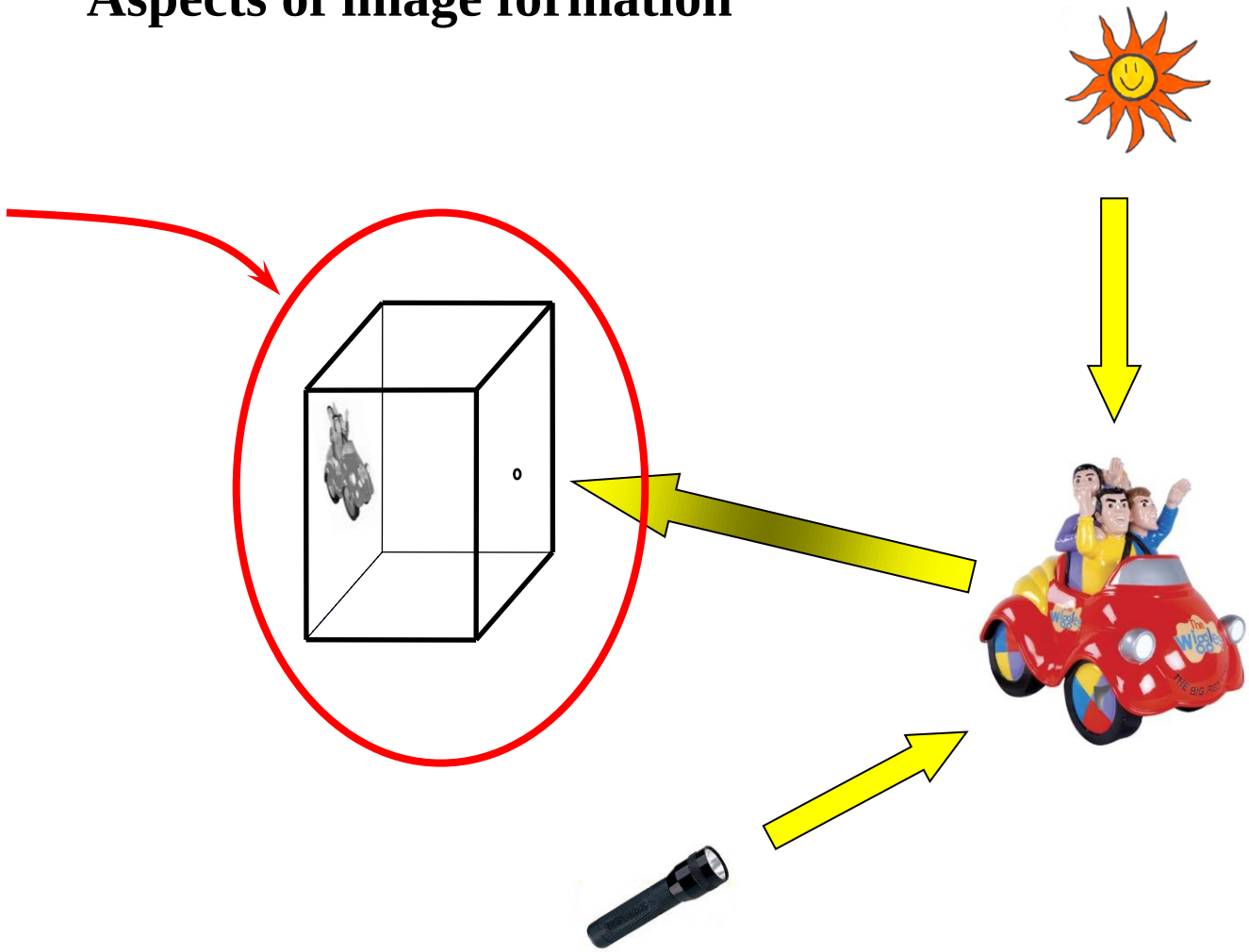
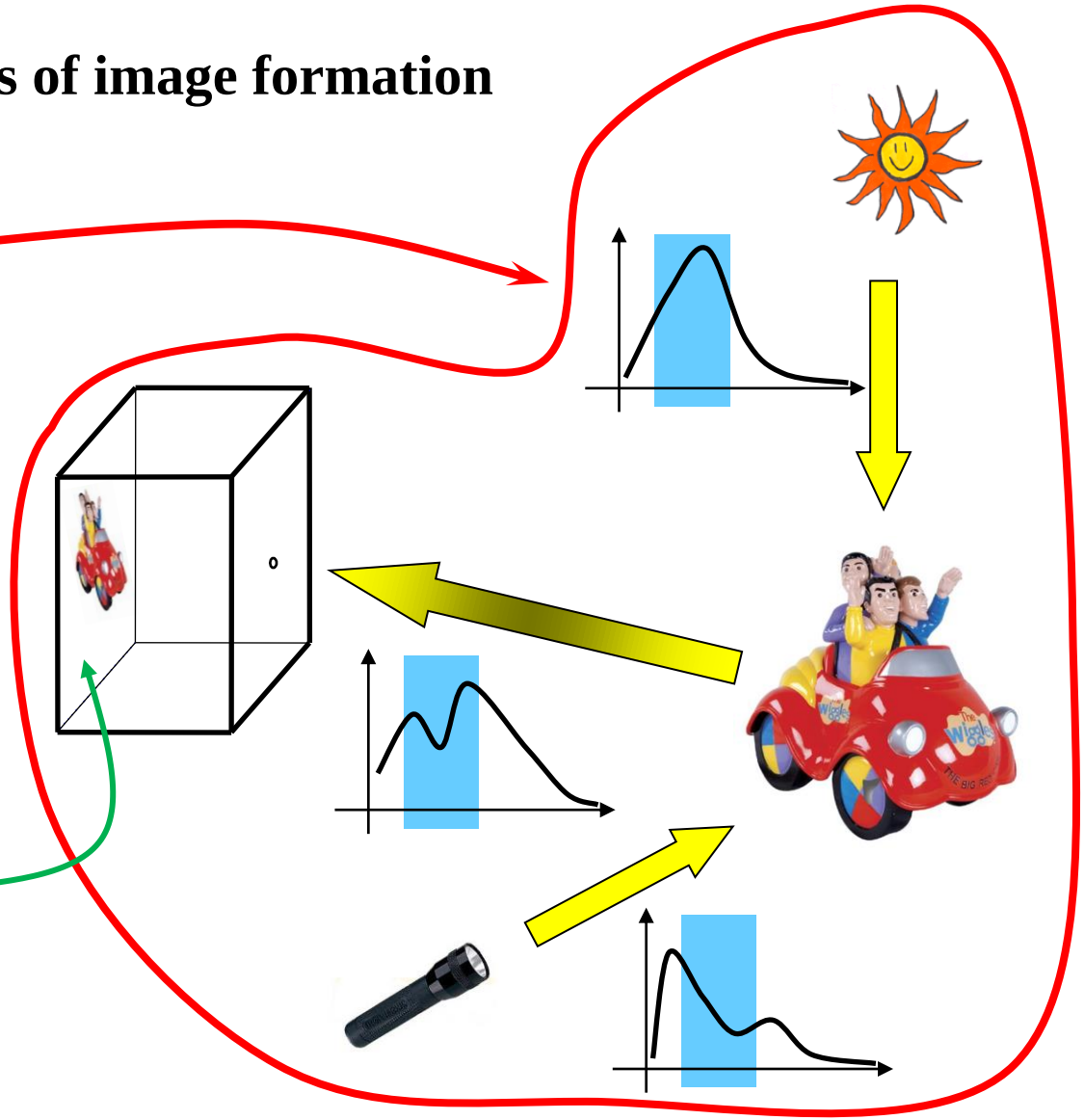


Image formation and representation

Aspects of image formation

1. Geometry
2. Optics
3. Radiometry
4. Color and colorimetry



$$I(x, y): R^2 \rightarrow R^*$$

Image formation and representation

Aspects of image formation

5. Sampling $\boldsymbol{I_s(i,j): Z^2 \rightarrow R}$
6. Quantization $\boldsymbol{I_s^q(i,j): Z^2 \rightarrow Z}$

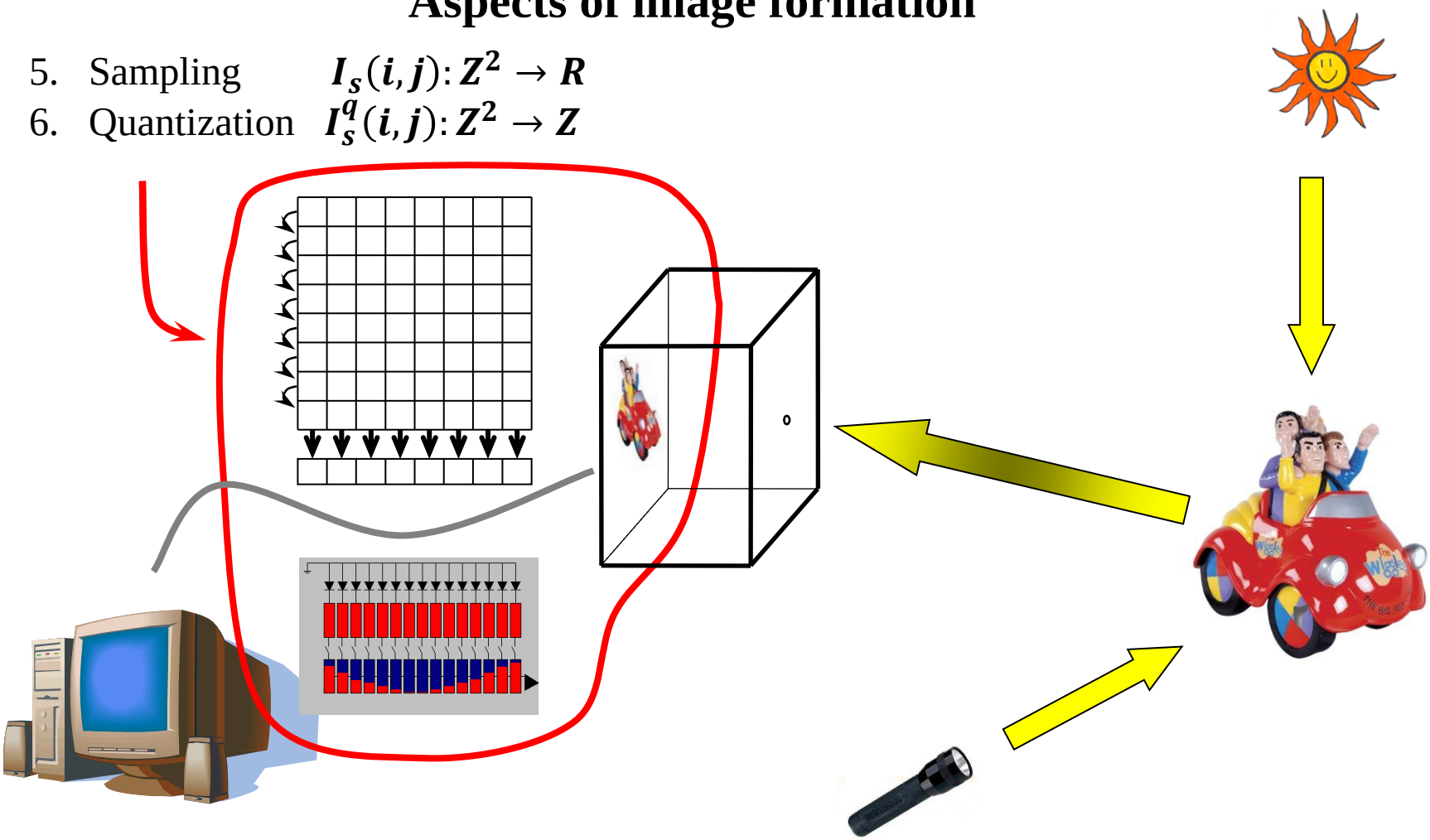


Image formation and representation

Aspects of image formation

5. Sampling $I_s(i,j): Z^2 \rightarrow R$
6. Quantization $I_s^q(i,j): Z^2 \rightarrow Z$ ➡ I^q (Stored/Represented value)

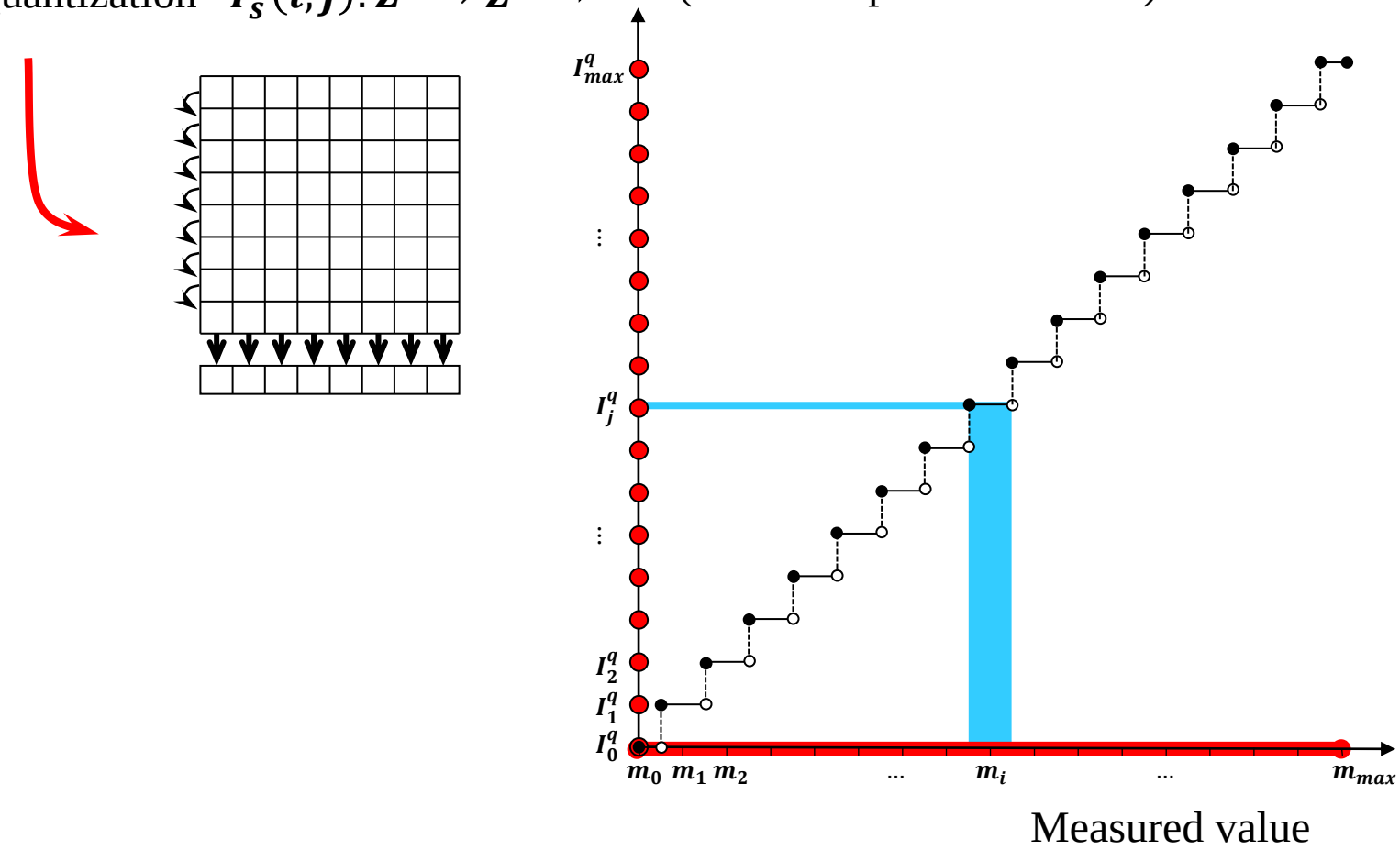
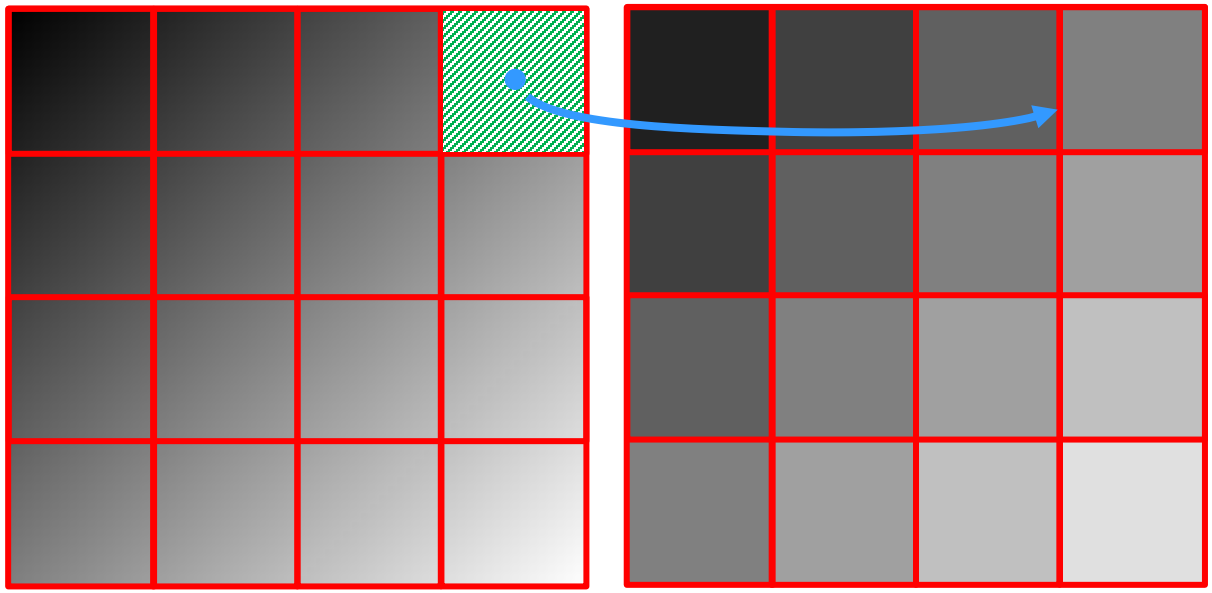


Image formation and representation

Aspects of image formation – Sampling and quantization

Point sampling: $I_s(i,j) = I(i \cdot \Delta x, j \cdot \Delta y)$

Integral sampling : $I_s(i,j) = \frac{\iint_{pixel} I(x,y)dxdy}{pixel\ area}$

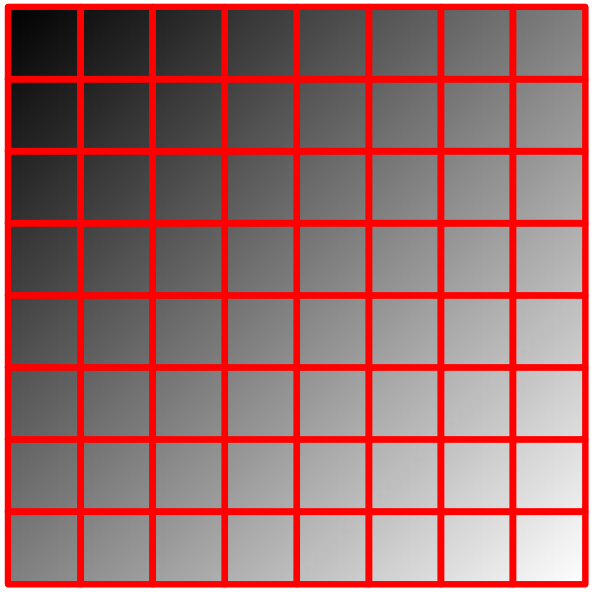


Camera input
(Continuous signal)

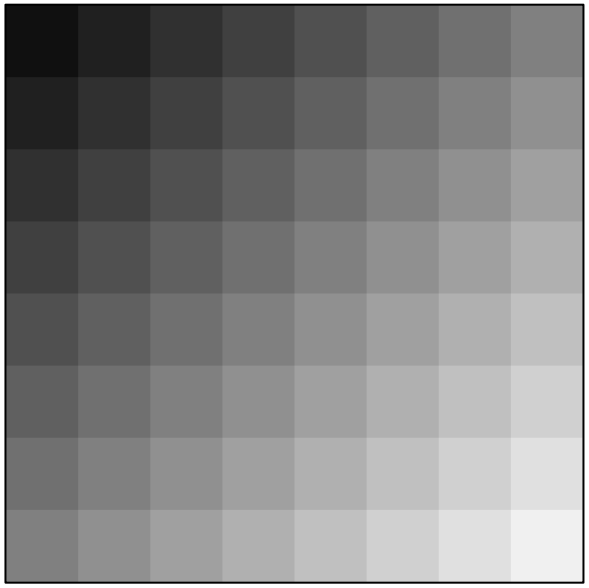
Sampled signal
(Discrete signal)

Image formation and representation

Aspects of image formation – Sampling and quantization



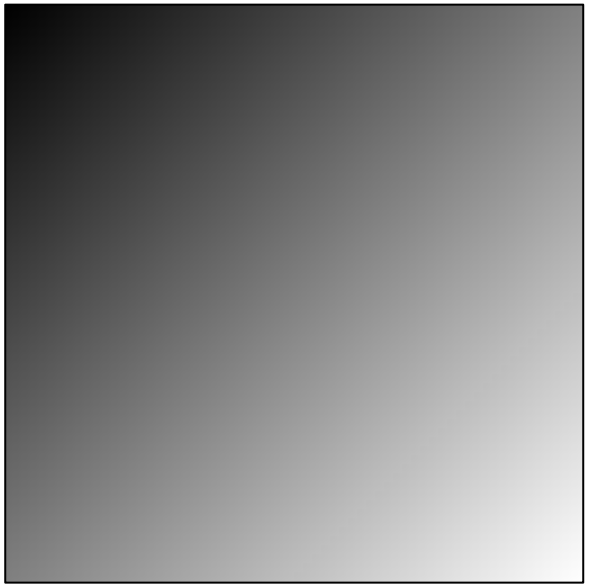
Camera input
(Continuous signal)



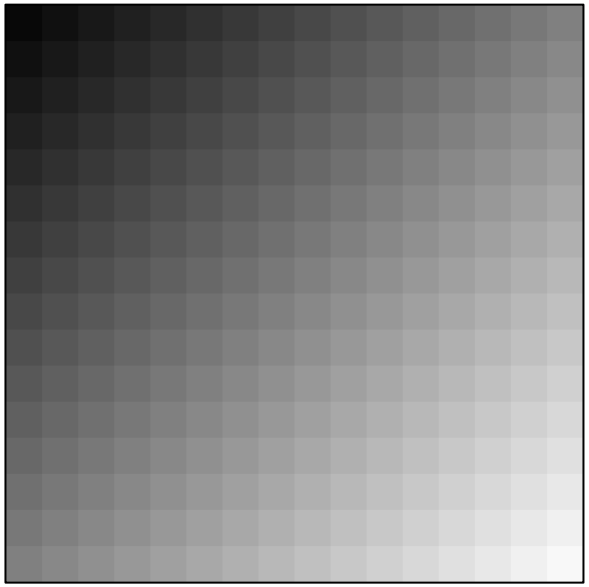
Sampled signal
(Discrete signal)

Image formation and representation

Aspects of image formation – Sampling and quantization



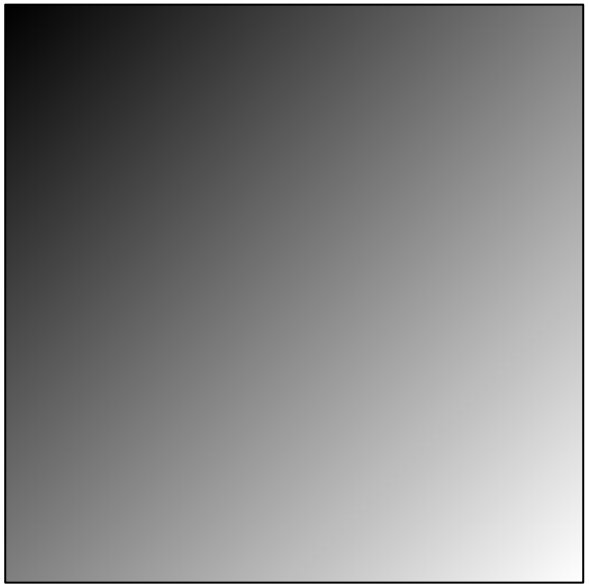
Camera input
(Continuous signal)



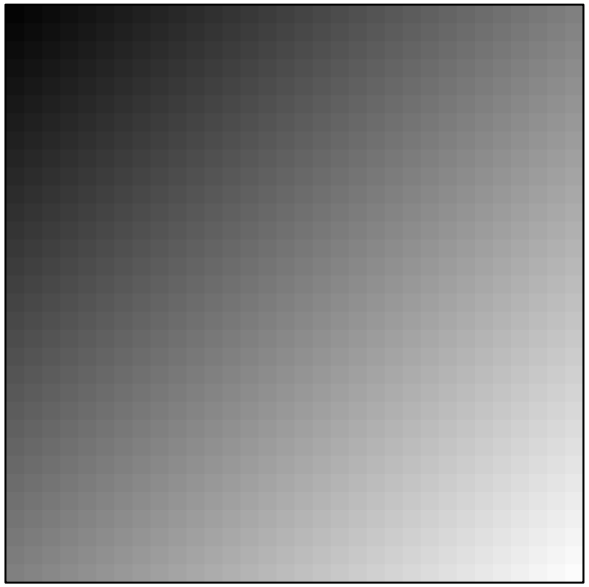
Sampled signal
(Discrete signal)

Image formation and representation

Aspects of image formation – Sampling and quantization



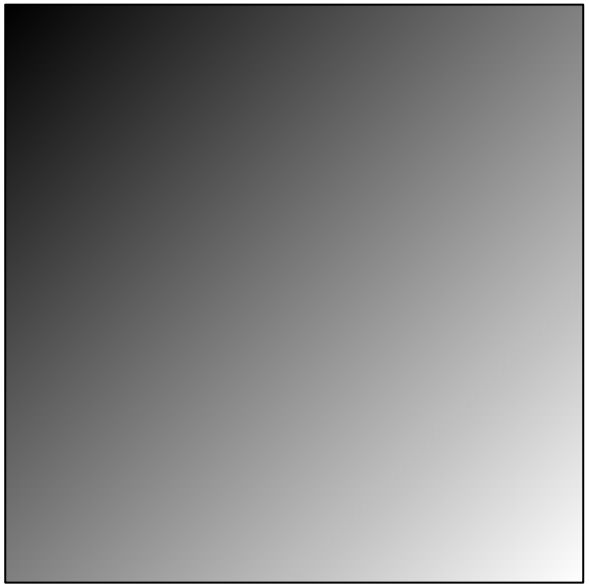
Camera input
(Continuous signal)



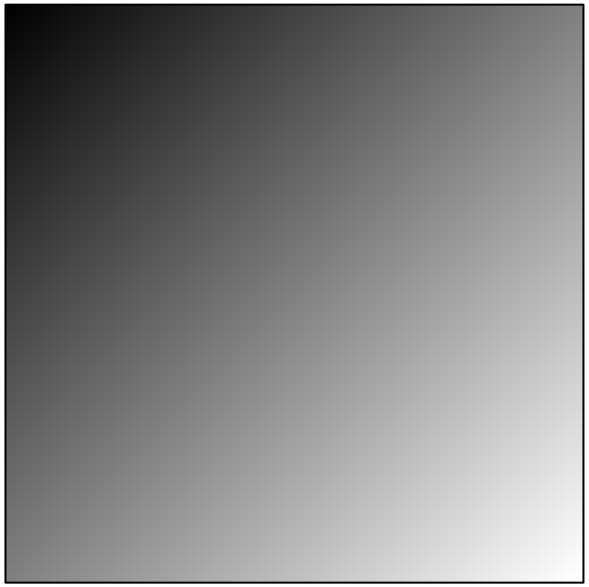
Sampled signal
(Discrete signal)

Image formation and representation

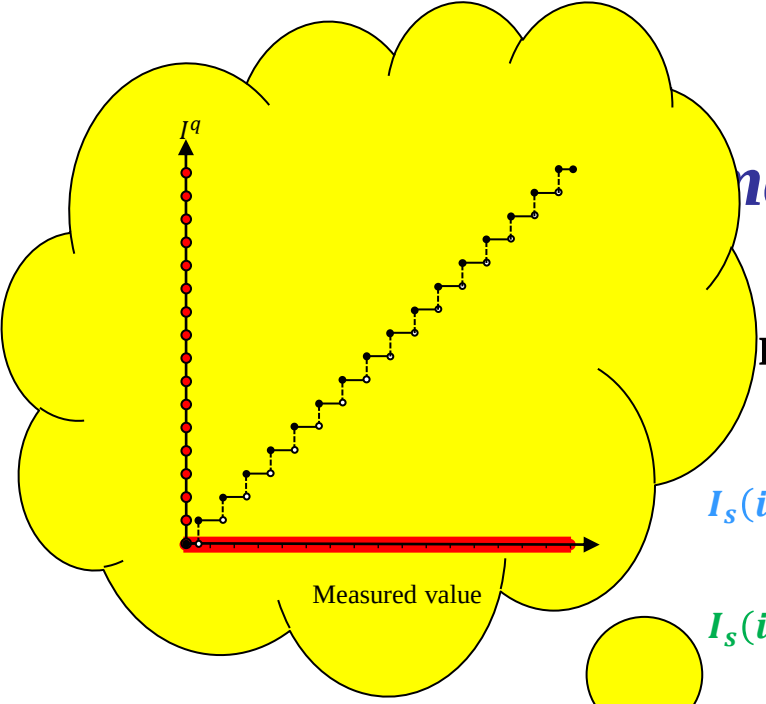
Aspects of image formation – Sampling and quantization



Camera input
(Continuous signal)



Sampled signal
(Discrete signal)



nation and representation

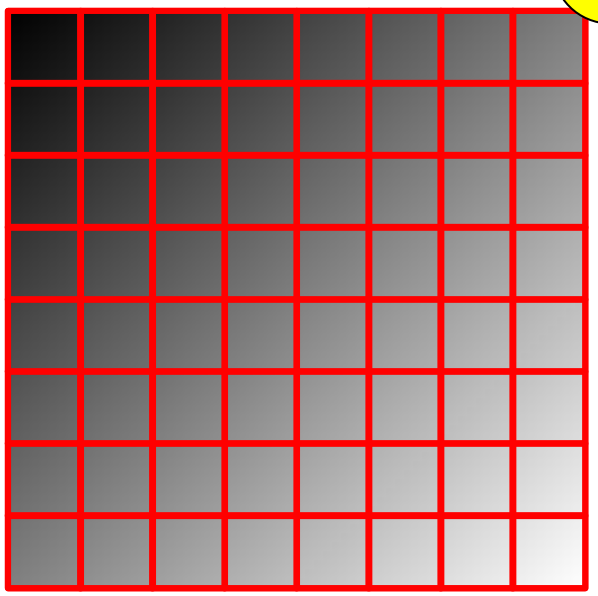
mation – Sampling and quantization

$$I_s(i,j) = I(i \cdot \Delta x, j \cdot \Delta y)$$

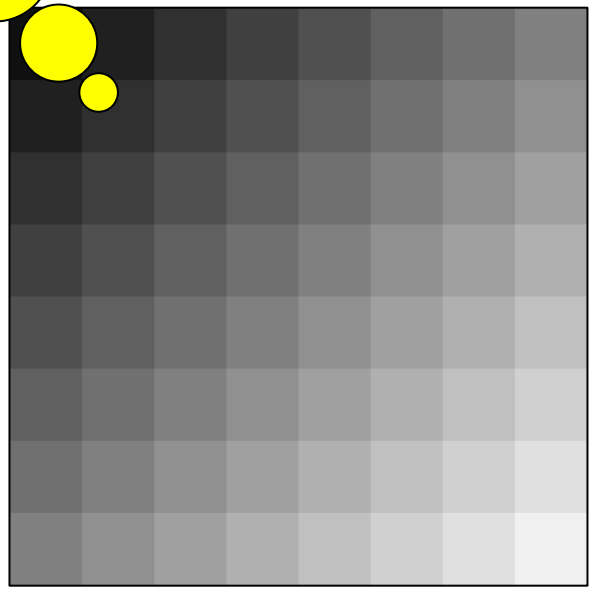
$$I_s^q(i,j) = I^q(I(i,j))$$

$$I_s(i,j) = \frac{\iint_{pixel} I(x,y)dxdy}{pixel\ area}$$

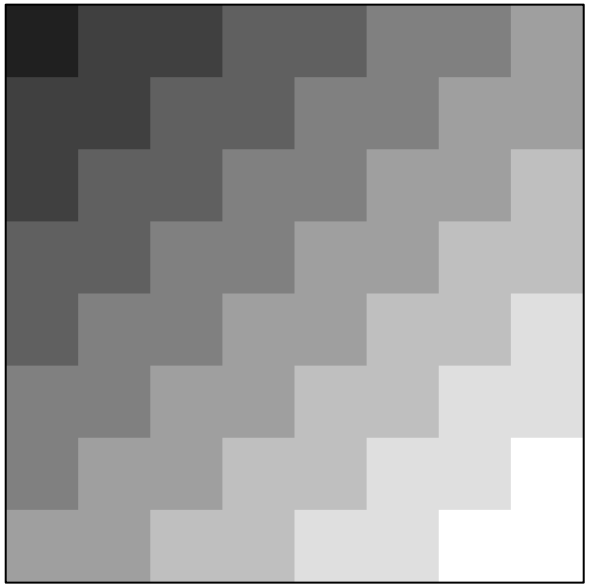
$$I_s^q(i,j) = I^q(I_s(i,j))$$



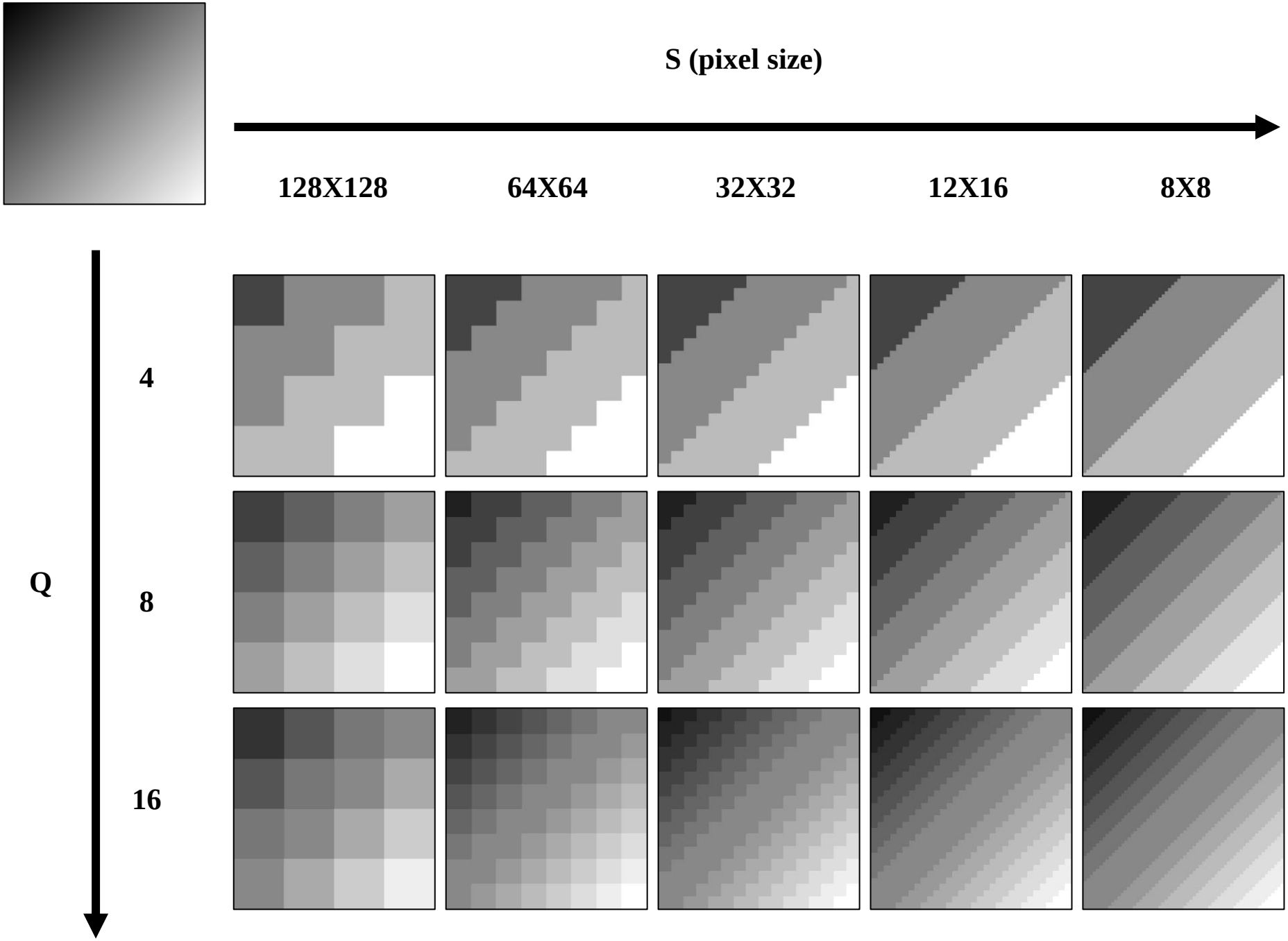
Camera input
(Continuous signal)



Sampled signal
(Discrete signal)



Quantized sampled signal
(Digital signal)


S (pixel size)
128X128
64X64
32X32
12X16
8X8
Q
4
8
16

S (pixel area)

32X32 **16X16** **8X8** **4X4** **2X2**

Q

4

8

16

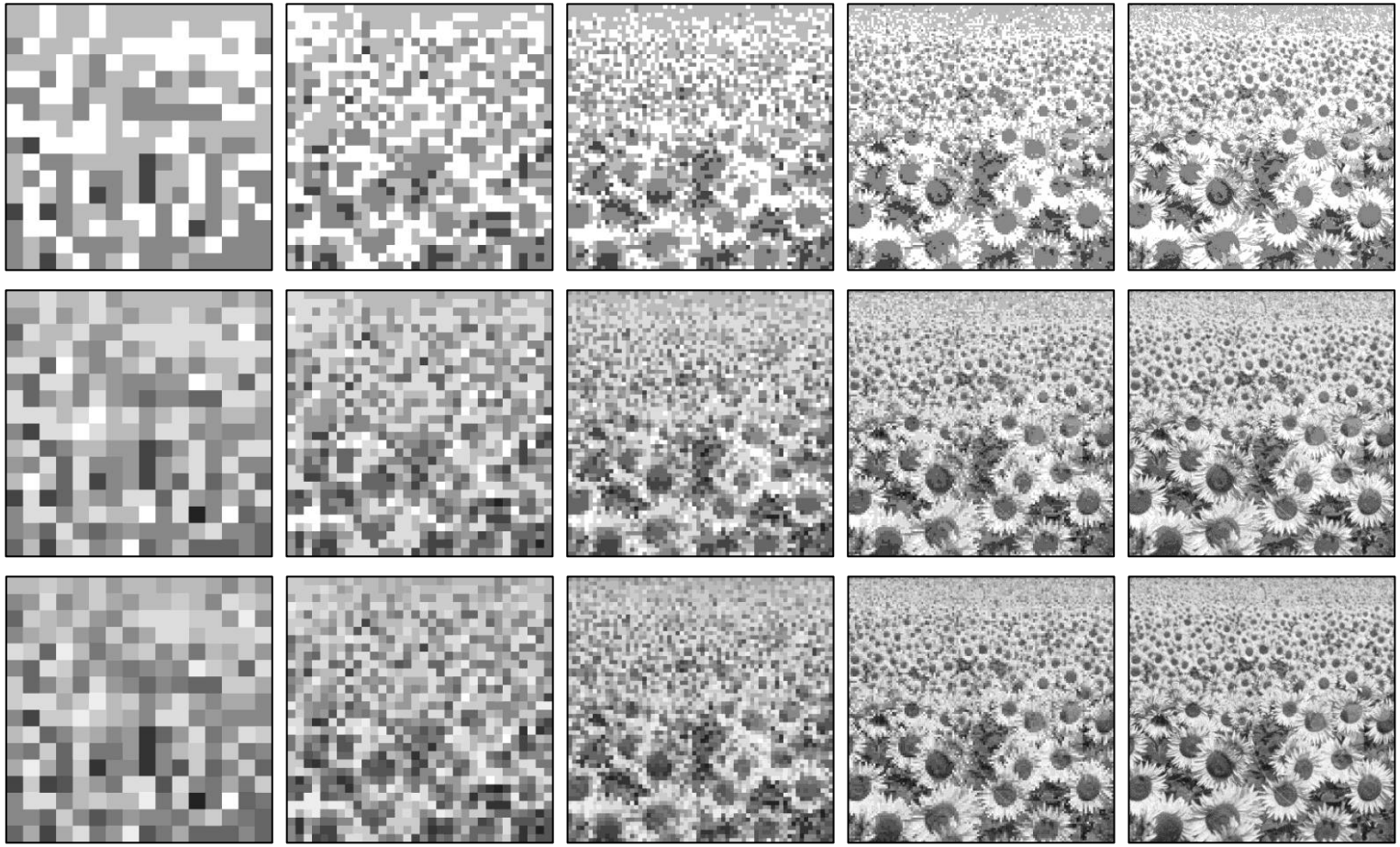


Image formation and representation

Aspects of image formation

1. Geometry
2. Optics
3. Radiometry
4. Color and colorimetry
5. Sampling
6. Quantization

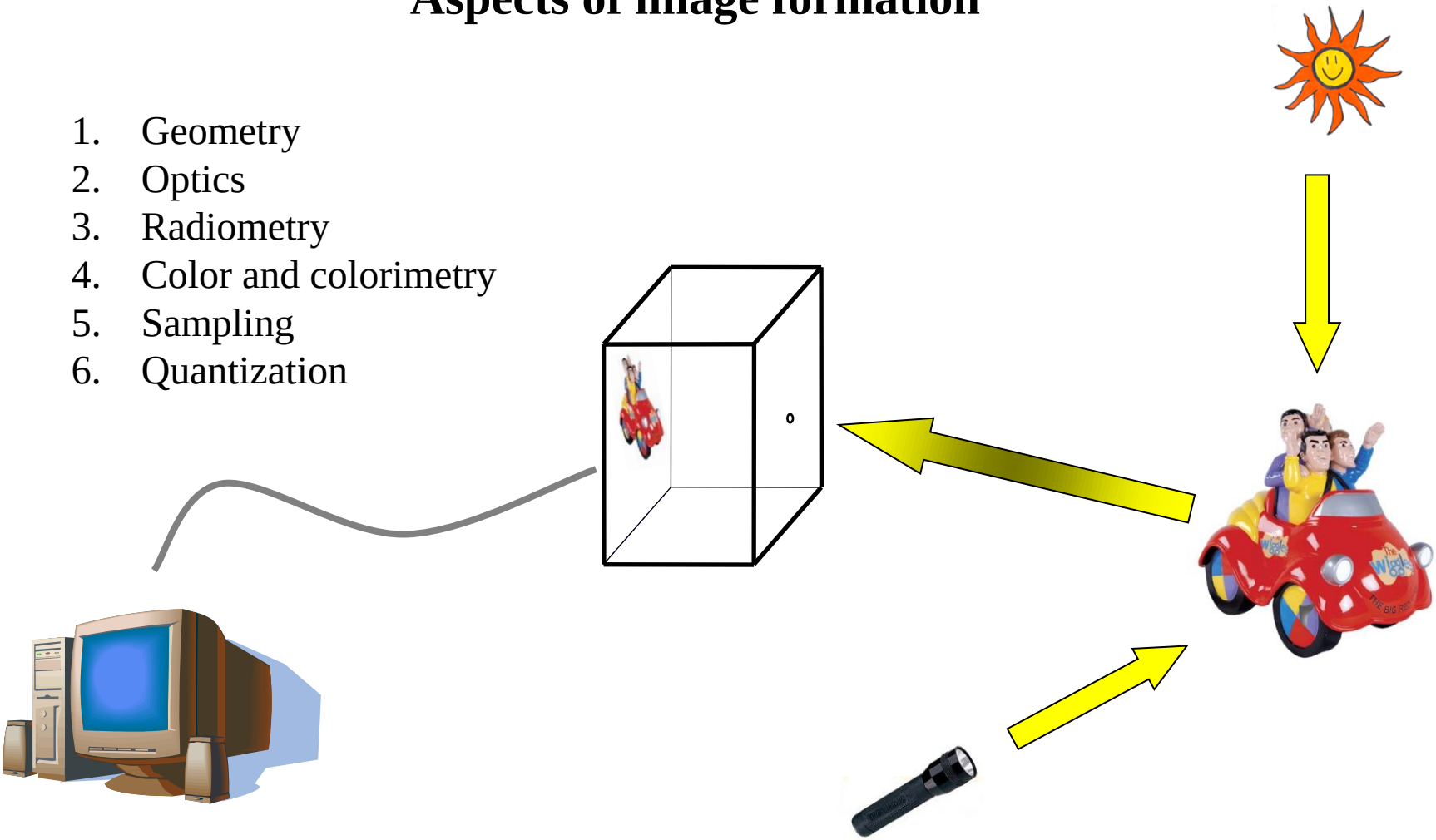


Image formation and representation

The issue of representation

Representation:

A *formal* system for making *explicit* certain types of *information* embedded in an *entity*.

Representational tradeoff:

Any particular representation makes certain information (or properties of the represented entities) explicit at the expense of other information (or properties) that is pushed into the background and may be quite hard to recover.

Therefore…

How information is represented greatly affects how easy it is to do certain things with it.

Image formation and representation

The issue of representation

Image formation and representation

Example: Representing numbers

1	**2**	**3**	**…**	**9**	**10**	**11**	**…**	**64**	**65**	**...**

Image formation and representation

Test your skills: What's in a representation…?

Image formation and representation

Image representation

(ignoring discretization and quantization)

$$I(x, y) : R^2 \rightarrow R$$

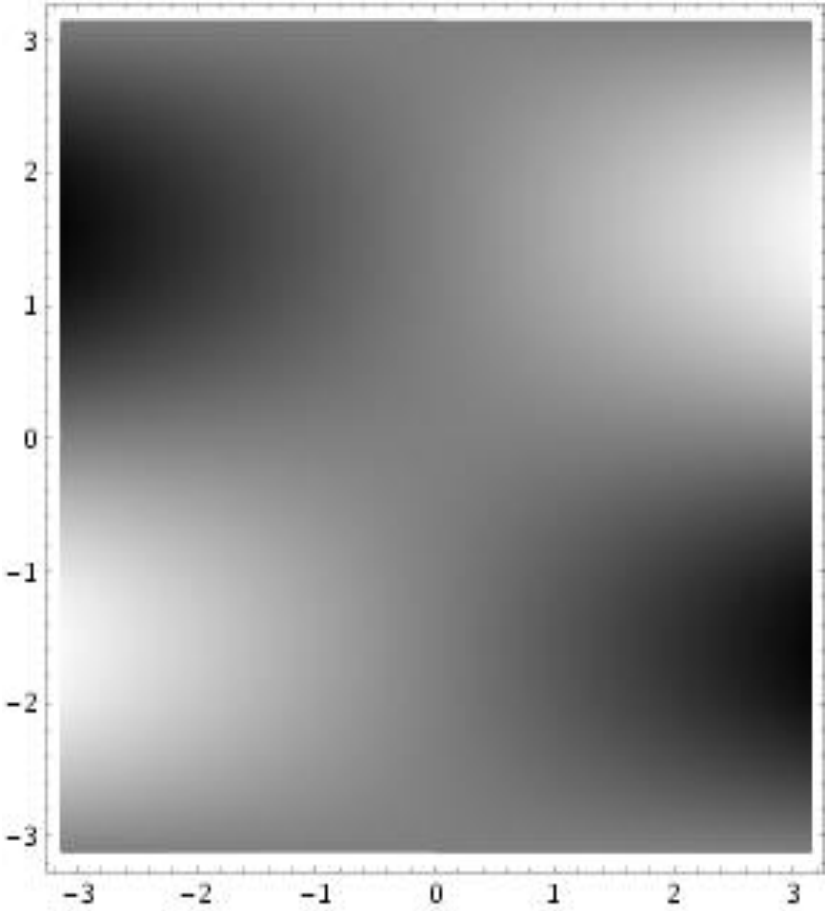
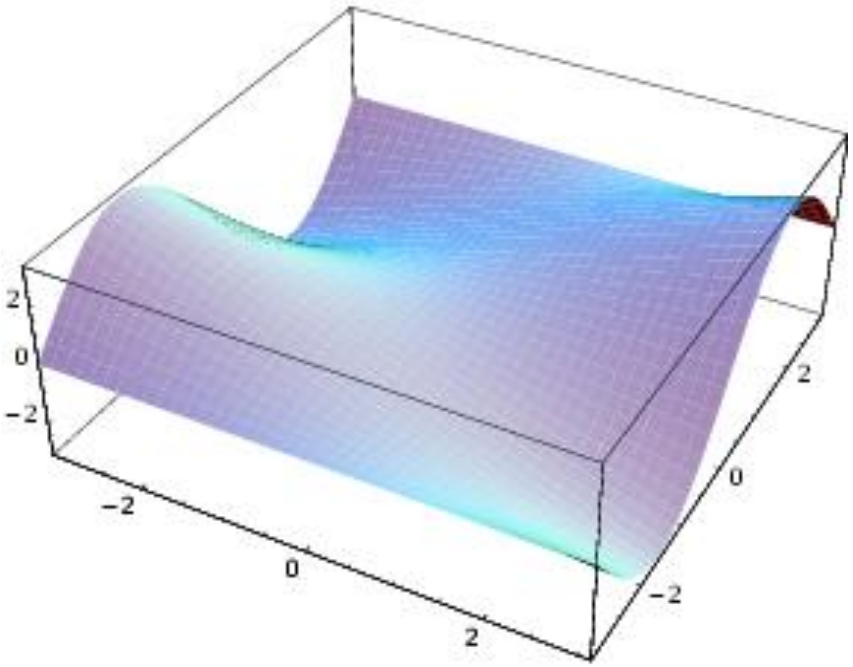


Image formation and representation

Image representation

(ignoring discretization and quantization)



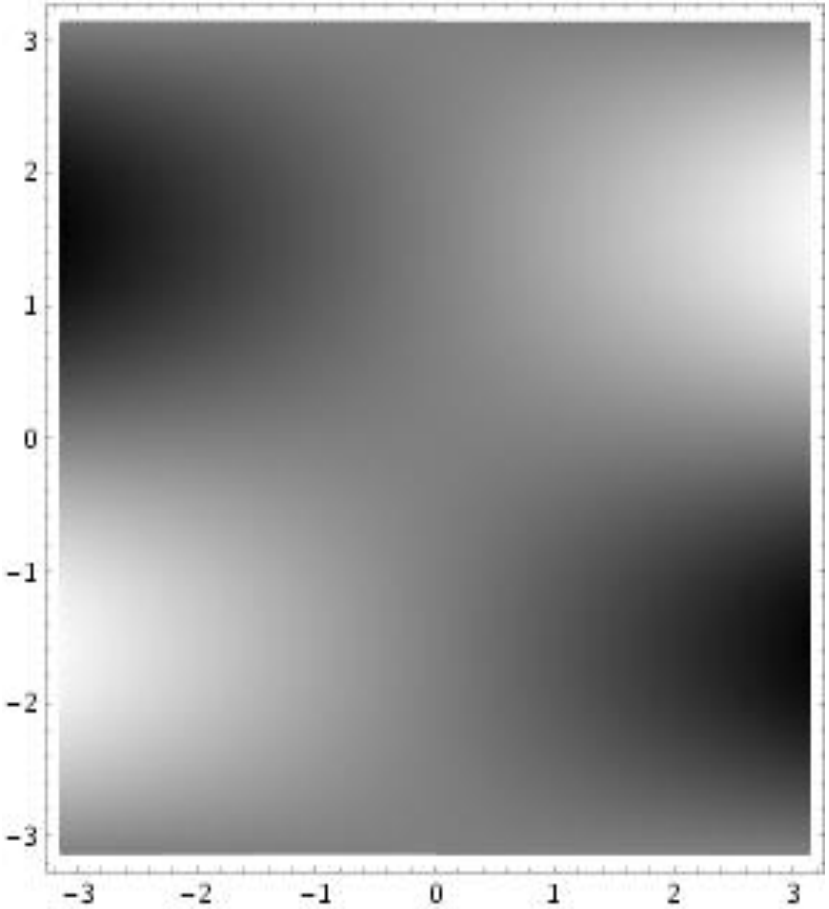
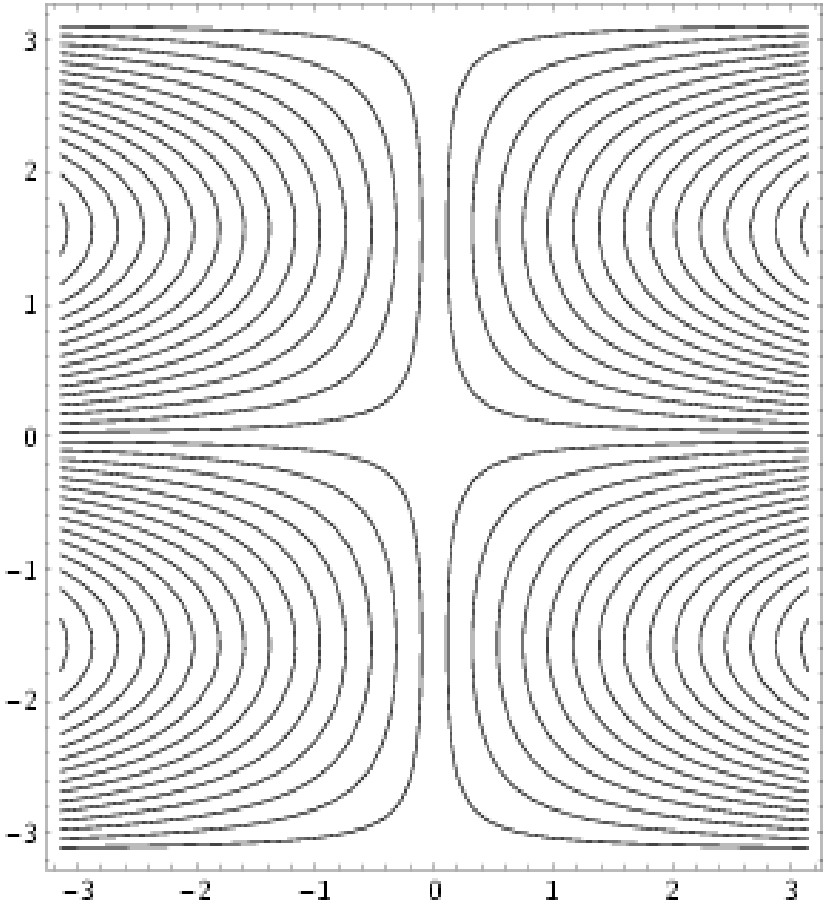


Image formation and representation

Image representation

(ignoring discretization and quantization)



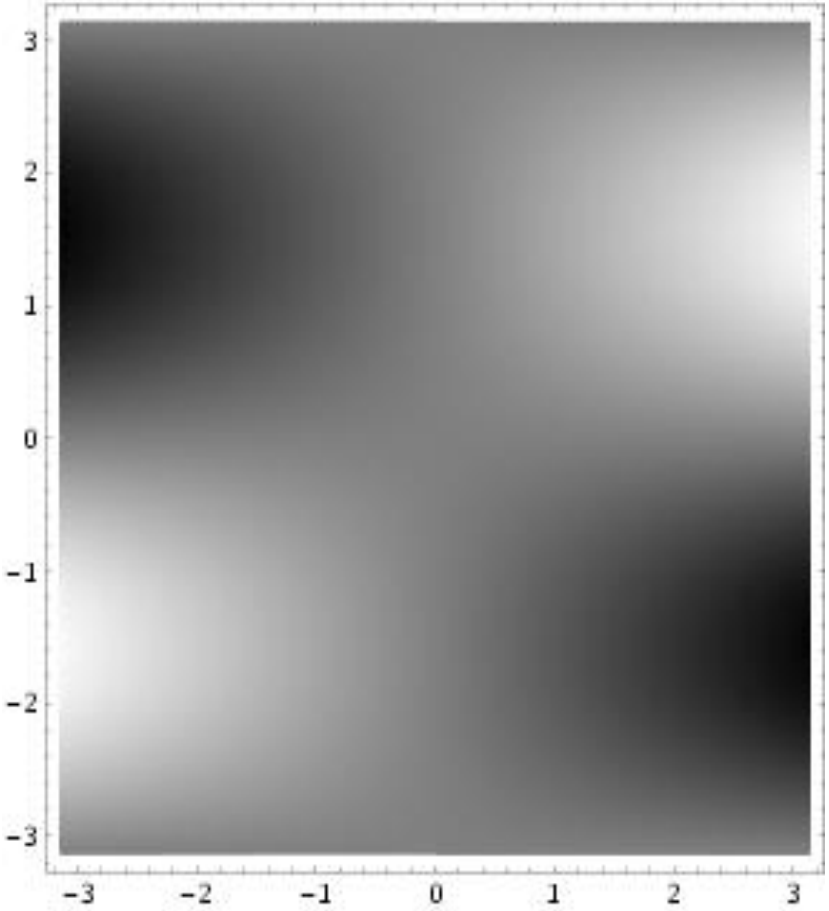


Image formation and representation

Image representation

M

$$\begin{pmatrix} \quad & \quad & \quad & \quad & \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad & \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad & \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad & \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad & \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad & \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad & \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad & \quad & \quad & \quad & \quad \end{pmatrix}$$

32	48	57	98	142	158	160	151
161	160	155	163	159	166	159	176
143	116	99	114	100	105	132	148
124	124	110	93	96	78	38	42
197	196	199	200	196	200	199	195
150	144	125	119	118	108	144	164
162	163	171	174	172	165	175	171
201	152	184	110	89	136	119	100
93	137	155	173	172	164	162	159

Image formation and representation

Image representation

$$M \longrightarrow \vec{v}$$

$$\begin{pmatrix}
32 & 48 & 57 & 98 & 142 & 158 & 160 & 151 \\
161 & 160 & 155 & 163 & 159 & 166 & 159 & 176 \\
143 & 116 & 99 & 114 & 100 & 105 & 132 & 148 \\
124 & 124 & 110 & 93 & 96 & 78 & 38 & 42 \\
197 & 196 & 199 & 200 & 196 & 200 & 199 & 195 \\
150 & 144 & 125 & 119 & 118 & 108 & 144 & 164 \\
162 & 163 & 171 & 174 & 172 & 165 & 175 & 171 \\
201 & 152 & 184 & 110 & 89 & 136 & 119 & 100 \\
93 & 137 & 155 & 173 & 172 & 164 & 162 & 159
\end{pmatrix}
\qquad
\begin{pmatrix} \vdots \end{pmatrix}$$

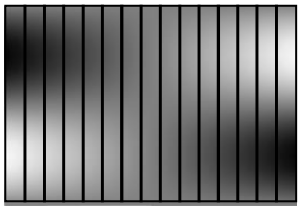


Image formation and representation

Image representation – Image space

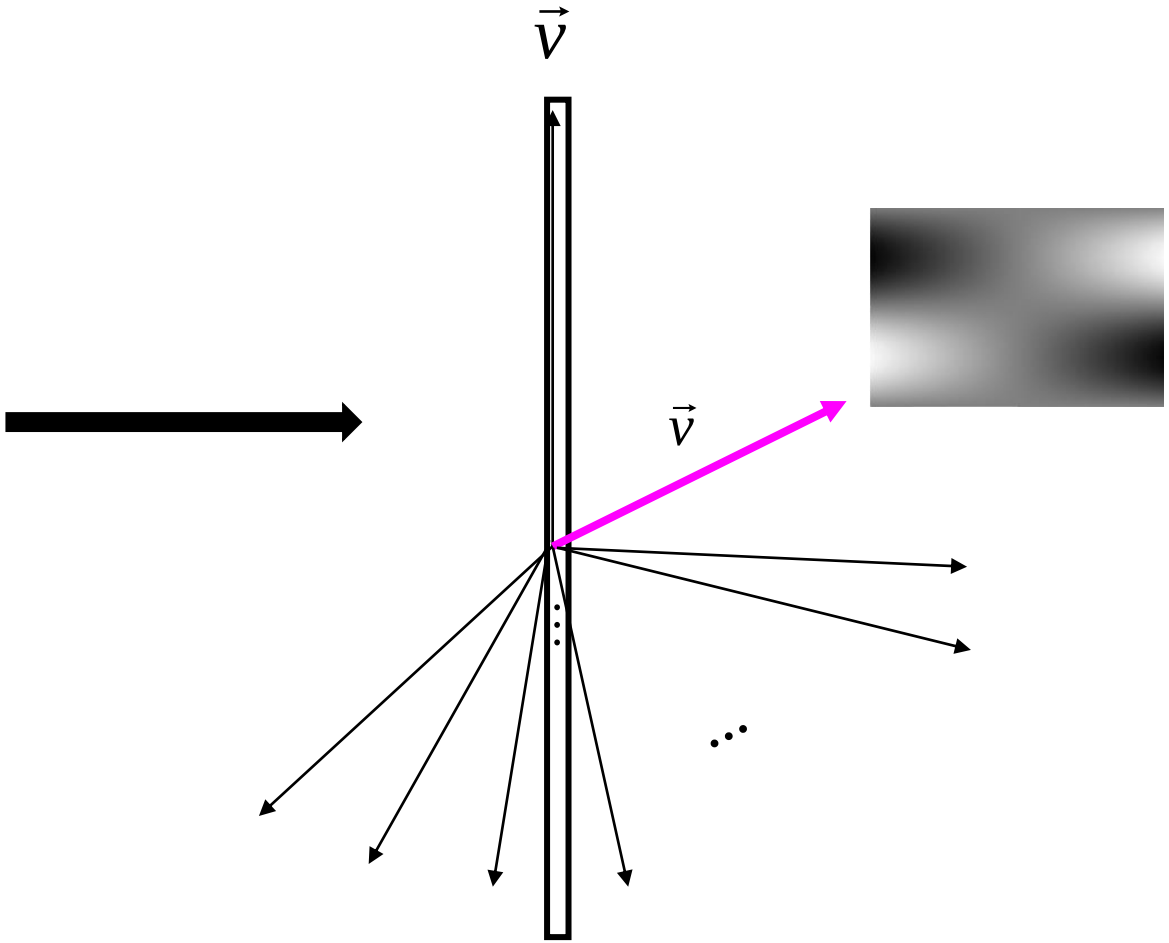


Image formation and representation

Image space – Is any point a "legitimate image"?

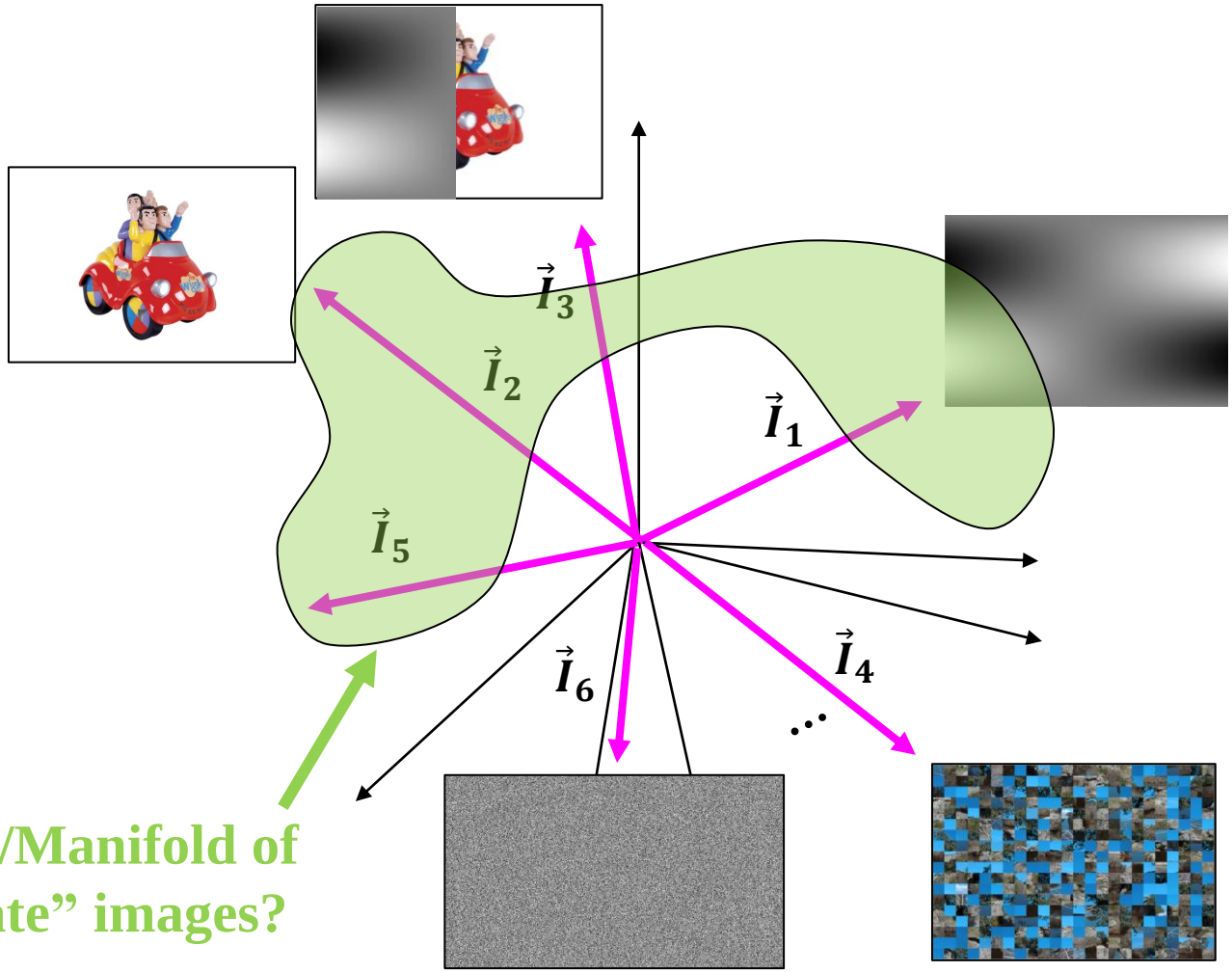


Image formation and representation

Aspects of image formation

1. Geometry
2. Optics
3. Radiometry
4. Color and colorimetry
5. Sampling
6. Quantization

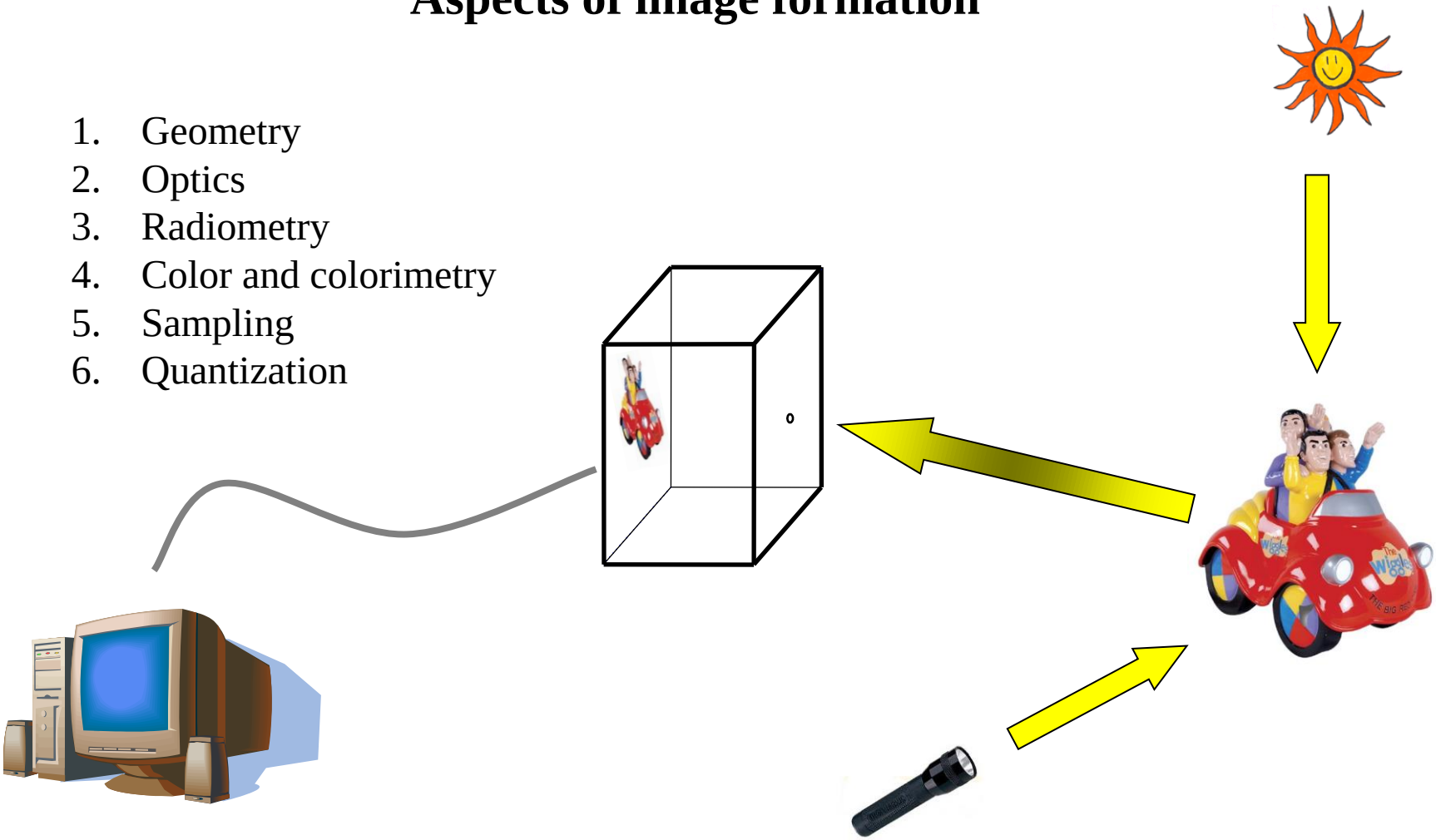


Image formation and representation

The geometry of image formation – The Pinhole camera model

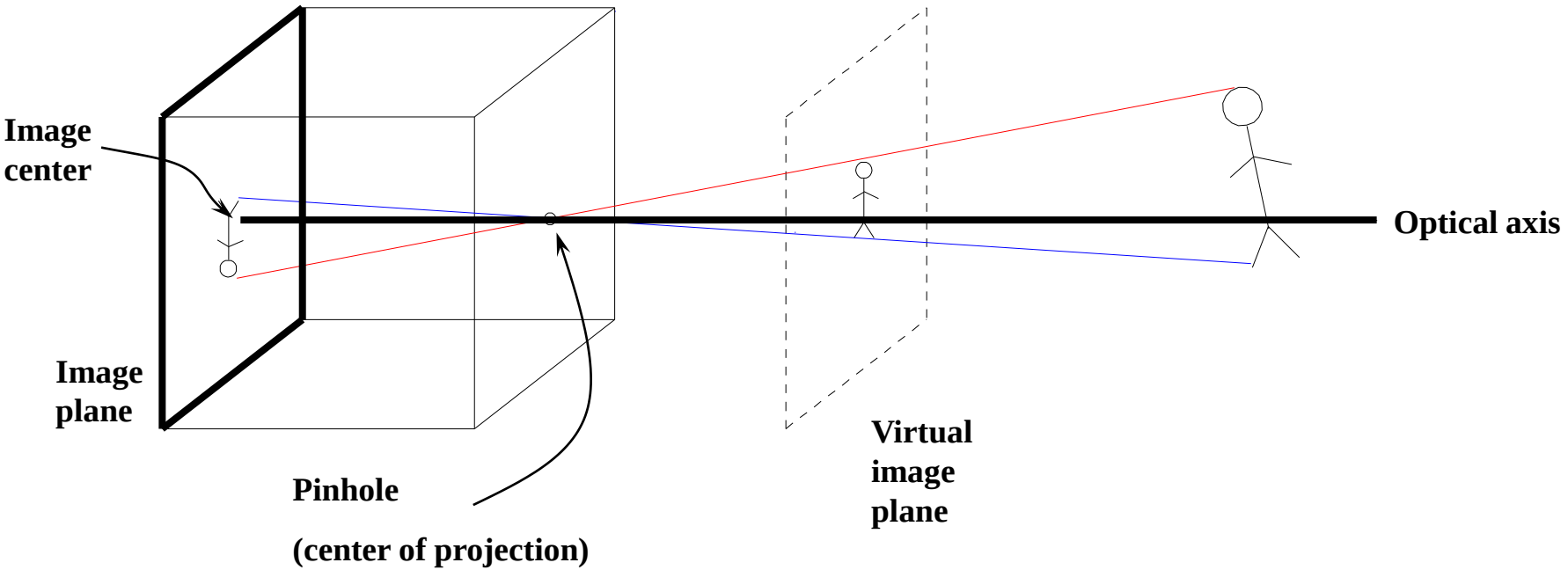
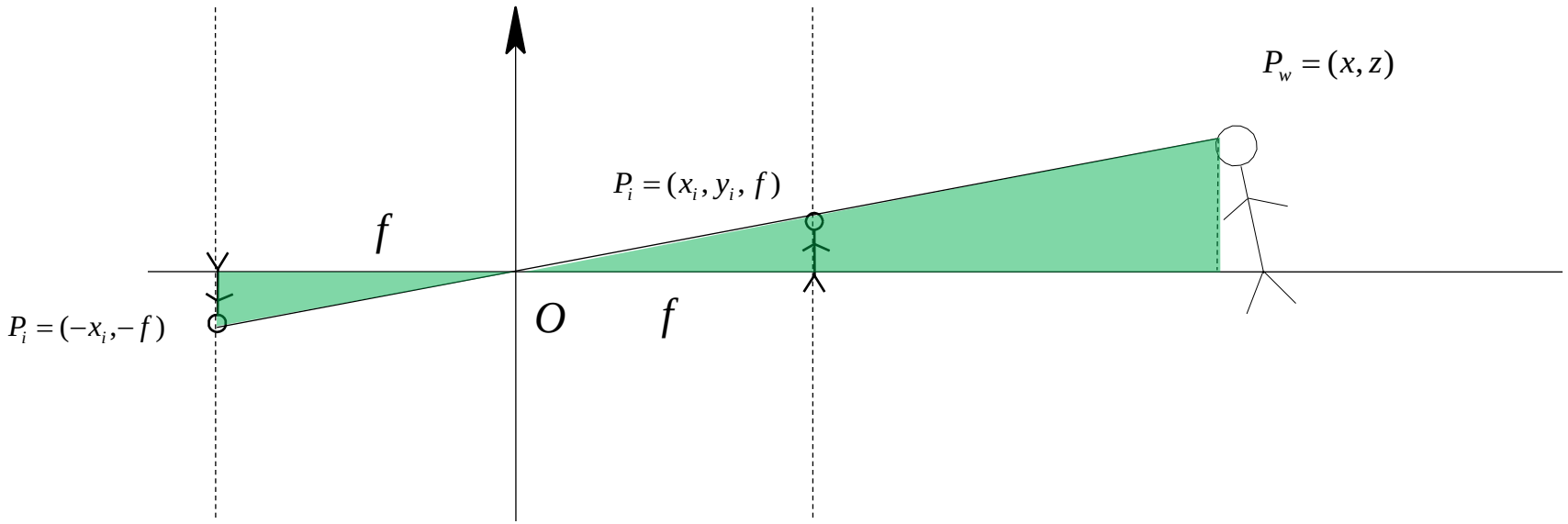


Image formation and representation

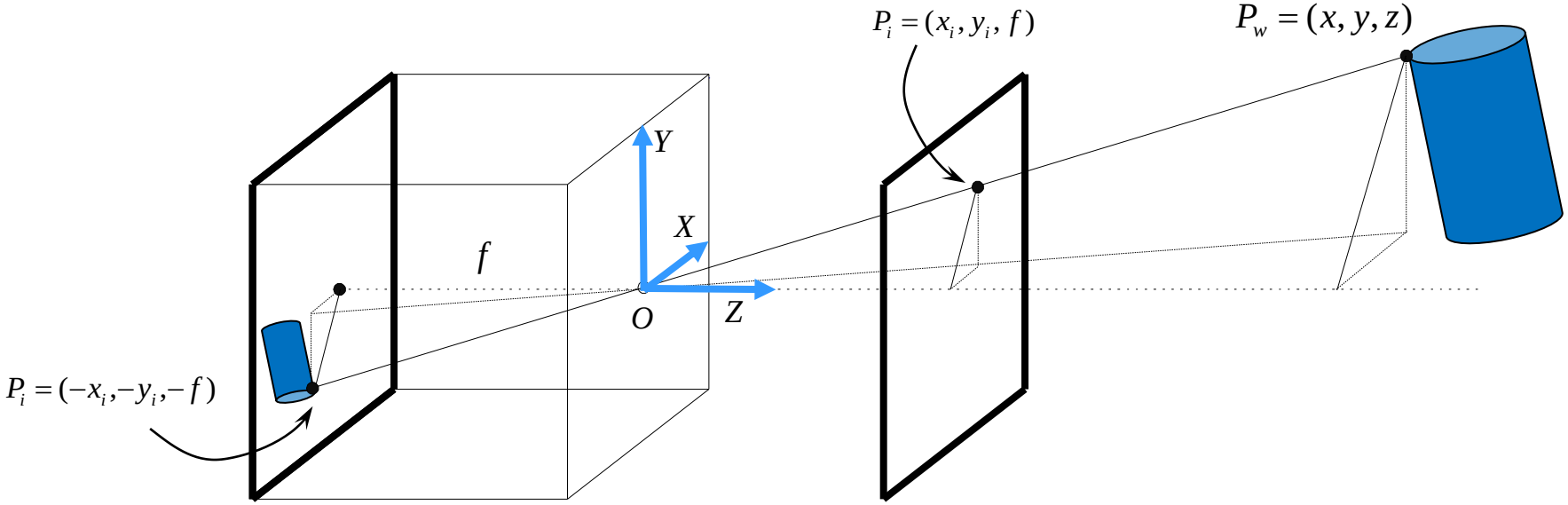
Pinhole camera model: Perspective projection



$$\frac{x_i}{f} = \frac{x}{z}$$

Image formation and representation

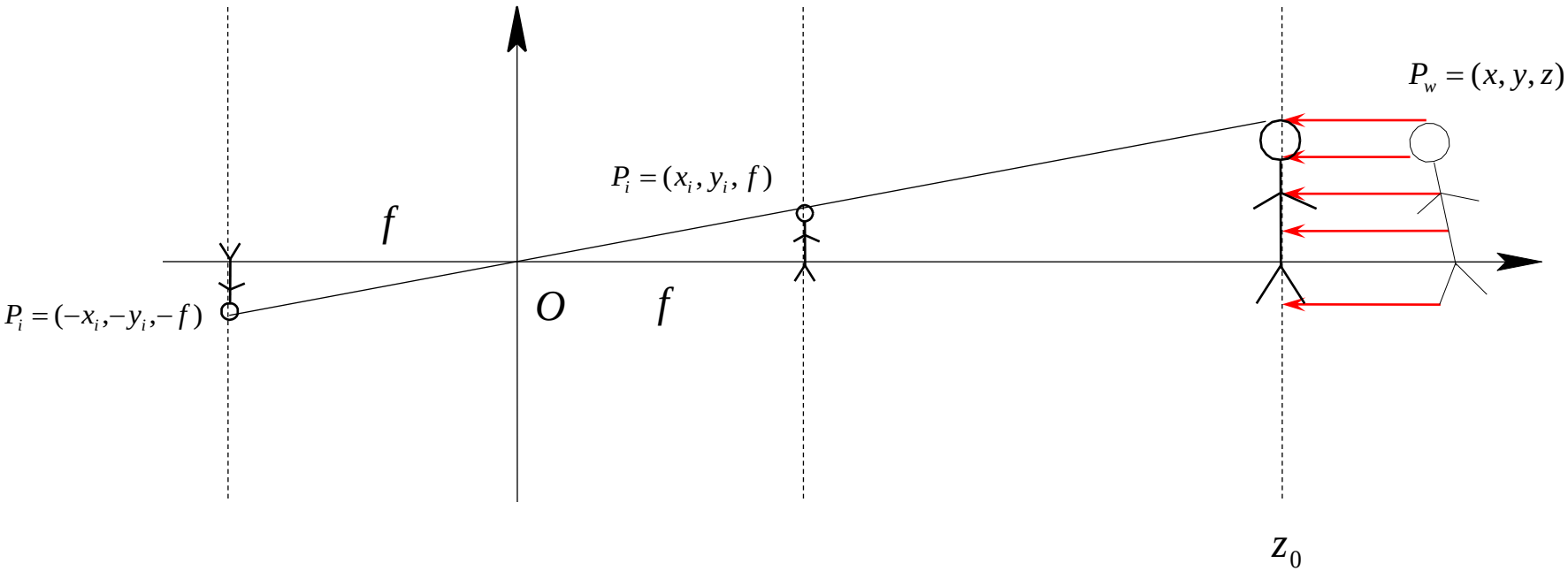
Pinhole camera model: Perspective projection



$$\overrightarrow{OP_i} = \lambda \cdot \overrightarrow{OP_w} \quad \Rightarrow \quad \lambda = \frac{x_i}{x} = \frac{y_i}{y} = \frac{f}{z} \quad \Rightarrow \quad \begin{cases} x_i = f \cdot \dfrac{x}{z} \\ y_i = f \cdot \dfrac{y}{z} \end{cases}$$

Image formation and representation

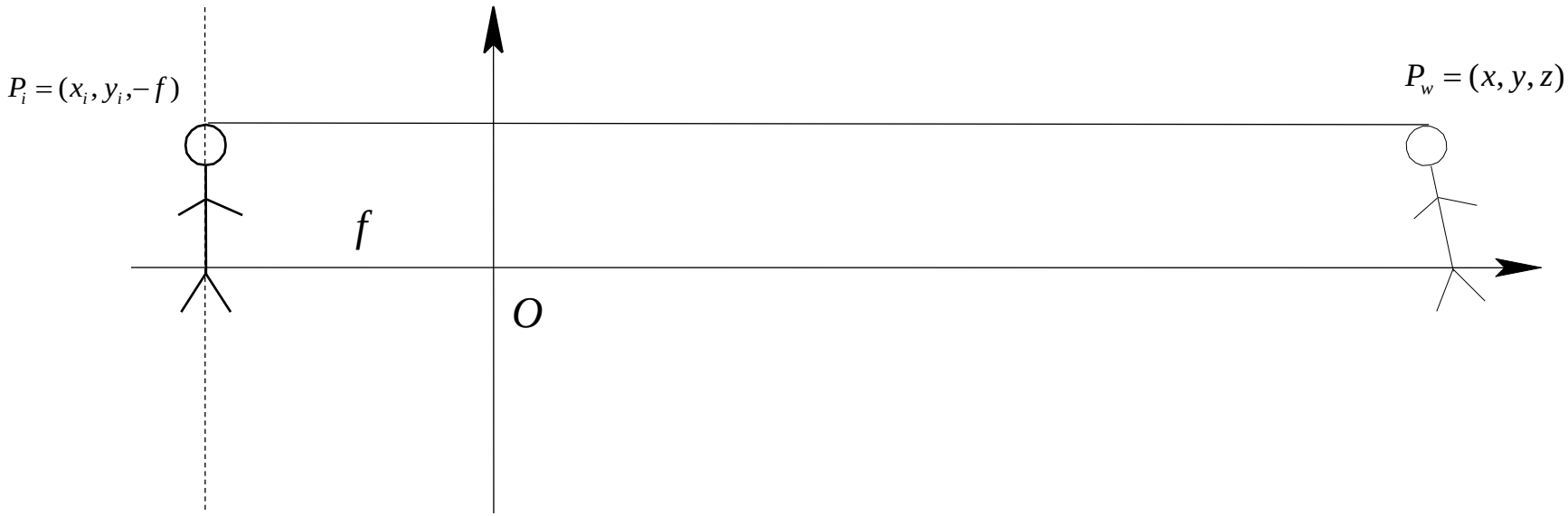
Linear approximation: Weak perspective projection



$$\begin{aligned} x_i &= s \cdot x \\ y_i &= s \cdot y \end{aligned} \Leftrightarrow \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{pmatrix} s & 0 \\ 0 & s \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Image formation and representation

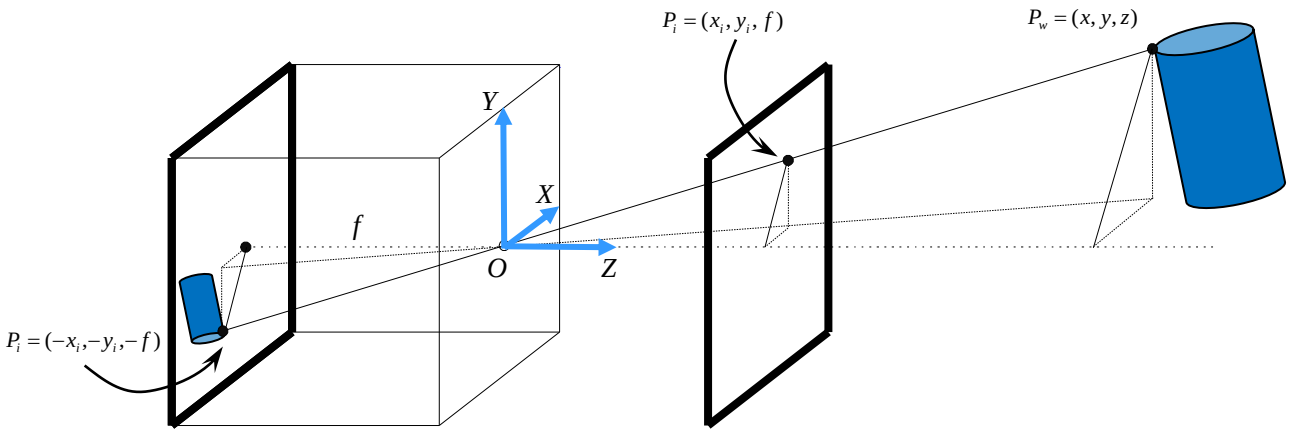
Linear approximation: Orthographic projection



$$\begin{aligned} x_i &= x \\ y_i &= y \end{aligned} \Leftrightarrow \begin{pmatrix} x_i \\ y_i \end{pmatrix} = I \begin{pmatrix} x \\ y \end{pmatrix}$$

Image formation and representation

Homogeneous coordinates

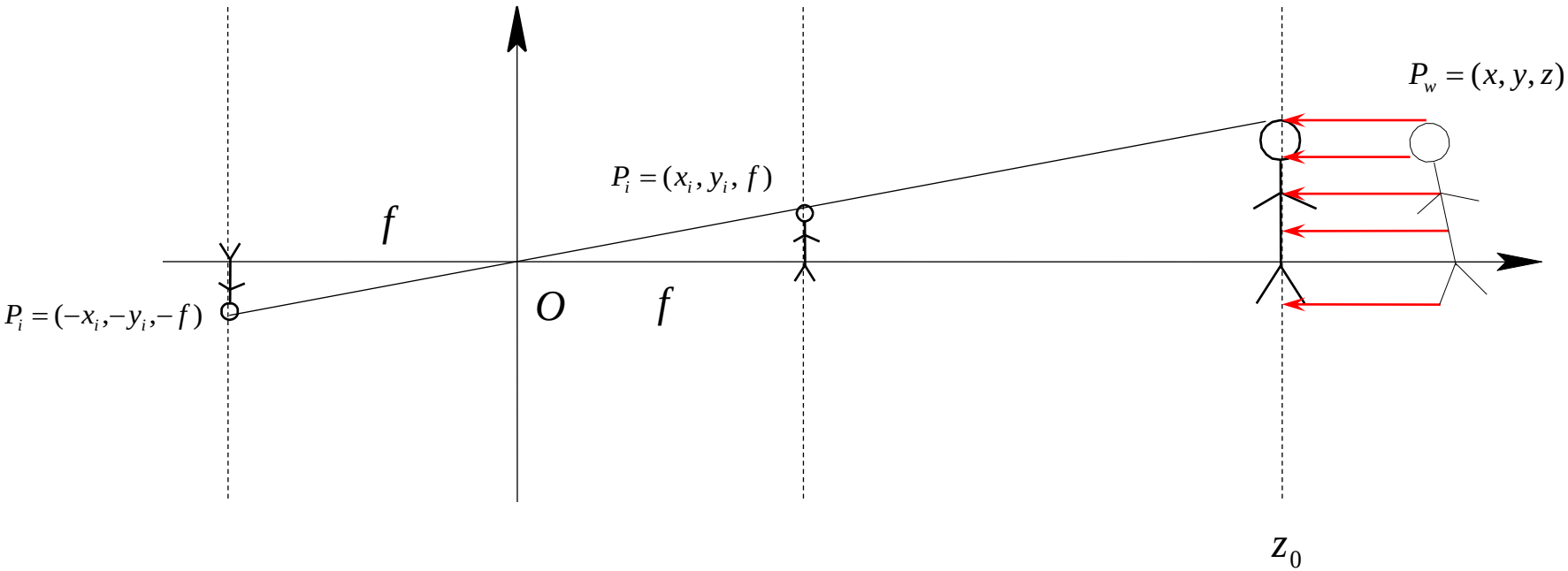


$$
\begin{aligned}
(x, y) &\Leftrightarrow [x \cdot w, y \cdot w, w] && w \neq 0 \\
(x, y, z) &\Leftrightarrow [x \cdot w, y \cdot w, z \cdot w, w] && w \neq 0
\end{aligned}
$$

most often w is set as $w = 1$

Image formation and representation

Weak perspective projection in homogeneous coordinates



$$\left.\begin{aligned} x_i &= s \cdot x \\ y_i &= s \cdot y \end{aligned}\right\} \leftrightarrow \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 & 0 \\ 0 & s & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Image formation and representation

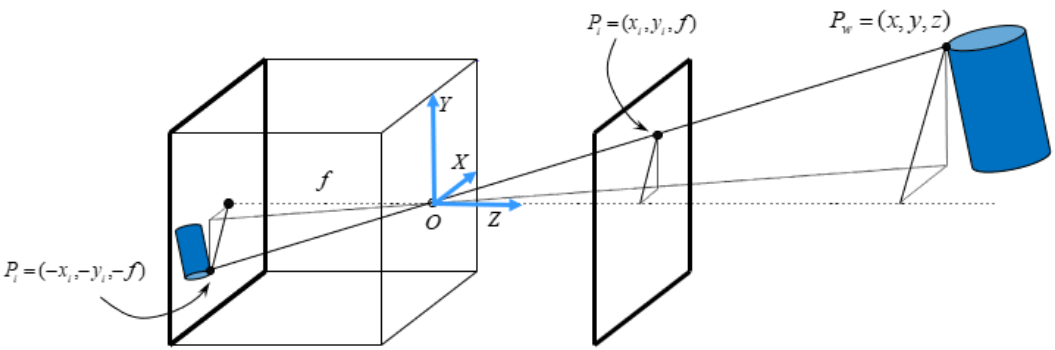
Perspective projection in homogeneous coordinates

$$\left.\begin{aligned} x_i &= f \cdot \frac{x}{z} \\ y_i &= f \cdot \frac{y}{z} \end{aligned}\right\} \leftrightarrow \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Image formation and representation

Perspective projection in homogeneous coordinates

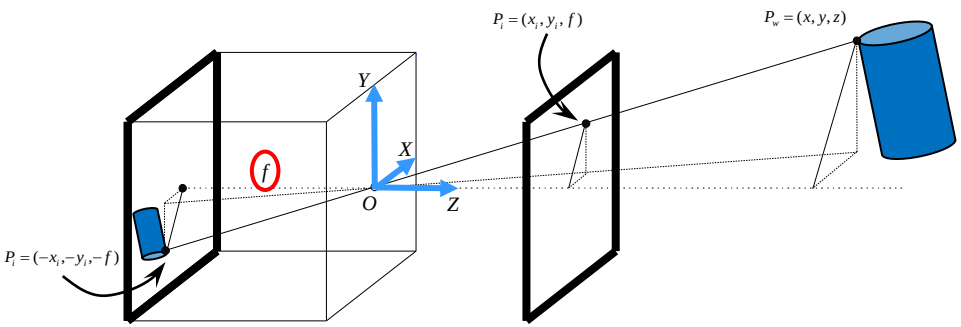
$$\left.\begin{aligned} x_i &= f \cdot \frac{x}{z} \\ y_i &= f \cdot \frac{y}{z} \end{aligned}\right\} \leftrightarrow \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



$$\left.\begin{aligned} x_i &= f \cdot \frac{x}{z} \\ y_i &= f \cdot \frac{y}{z} \end{aligned}\right\} \leftrightarrow \begin{pmatrix} x_i \\ y_i \\ f \end{pmatrix} = \begin{bmatrix} x \\ y \\ z \\ z/f \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/f & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Image formation and representation

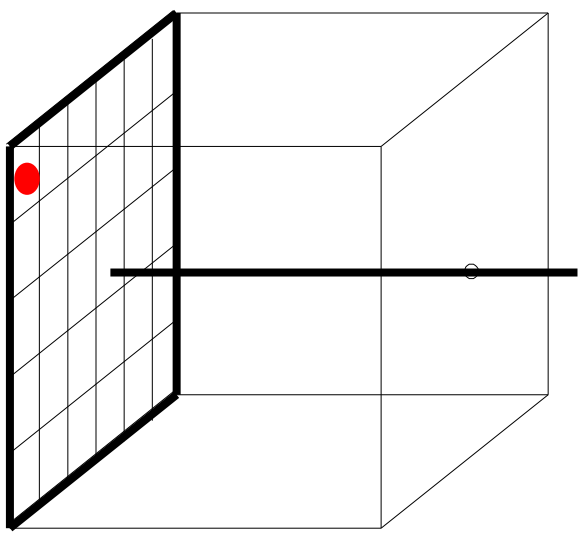
Camera *intrinsic* parameters



$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$x_i = S_x f \cdot \frac{x}{z} = \alpha \frac{x}{z} + x_0$$

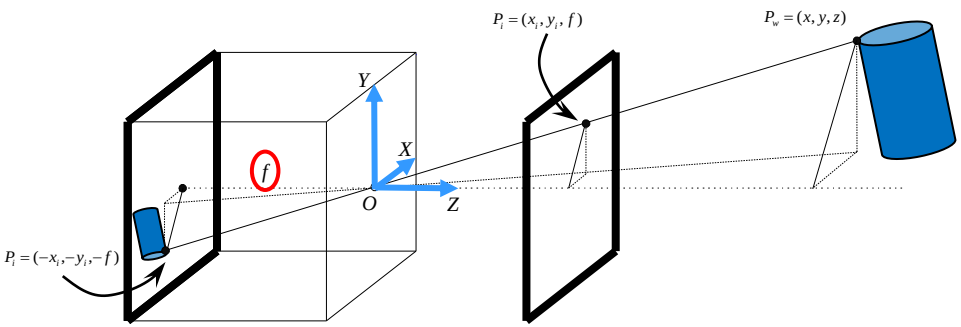
$$y_i = S_y f \cdot \frac{y}{z} = \beta \frac{y}{z} + y_0$$



$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} \alpha \cdot x + x_0 z \\ \beta \cdot y + y_0 z \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} \alpha & 0 & x_0 & 0 \\ 0 & \beta & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\text{The calibration matrix } \mathbf{K}} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Image formation and representation

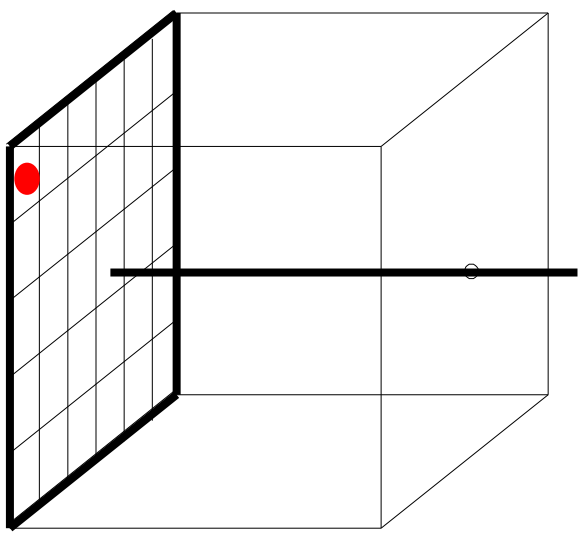
Camera *intrinsic* parameters



$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$x_i = S_x f \cdot \frac{x}{z} = \alpha \frac{x}{z} + x_0$$

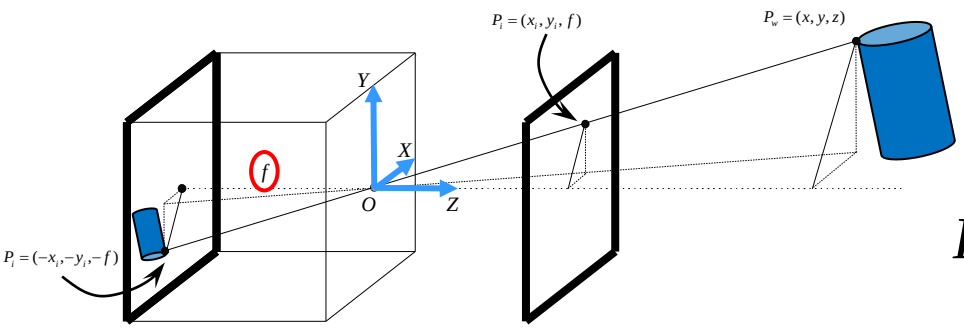
$$y_i = S_y f \cdot \frac{y}{z} = \beta \frac{y}{z} + y_0$$



$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ z \\ 1 \end{bmatrix} = \underbrace{\left[\begin{array}{ccc|c} \alpha & 0 & x_0 & 0 \\ 0 & \beta & y_0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right]}_{\text{The calibration matrix } \mathbf{K}} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Image formation and representation

Camera calibration = Recovering the parameters of an unknown camera



$$P_i^{\,j} = \begin{bmatrix} \alpha & 0 & x_0 & 0 \\ 0 & \beta & y_0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = K \cdot P_w^{\,j}$$

- Measure enough pairs $\left(P_i^{\,j}; P_w^{\,j}\right) \quad j = 1..N$
- Obtain enough linear equations in the parameters of K
- Solve for the parameters of K

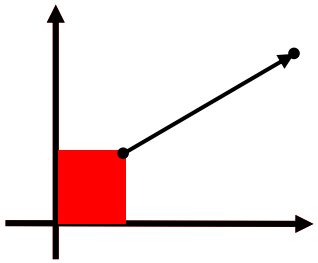
Problem : measuring $P_i^{\,j}$ is easy, but how can we measure $P_w^{\,j}$?

Image formation and representation

2D Transformations

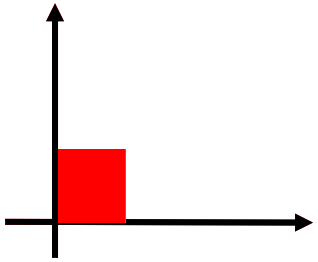
Translation $P_t = P_s + \vec{t}$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} x_s \\ y_s \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Rotation $P_t = R \cdot P_s$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{pmatrix} \begin{pmatrix} x_s \\ y_s \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Scaling $P_t = s\, P_s$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = s \begin{pmatrix} x_s \\ y_s \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$

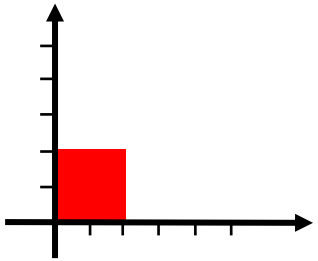
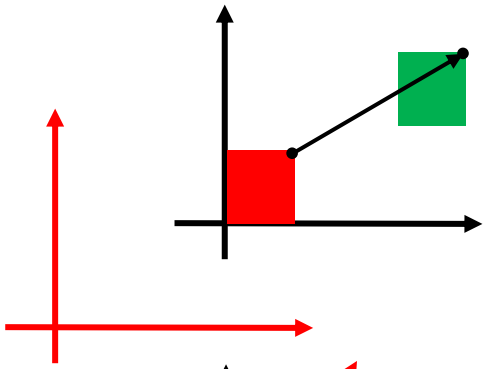


Image formation and representation

2D Transformations

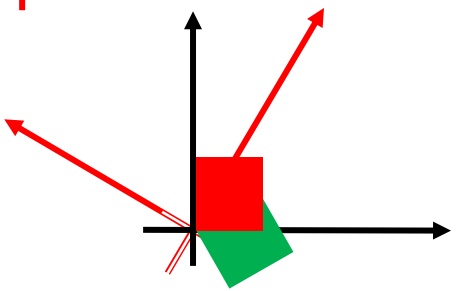
Translation $P_t = P_s + \vec{t}$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} x_s \\ y_s \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Rotation $P_t = R \cdot P_s$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{pmatrix} \begin{pmatrix} x_s \\ y_s \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Scaling $P_t = s\, P_s$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = s \begin{pmatrix} x_s \\ y_s \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$

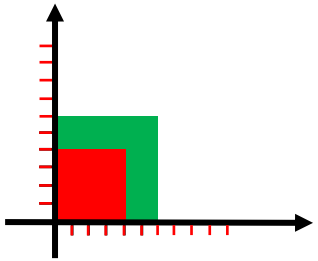
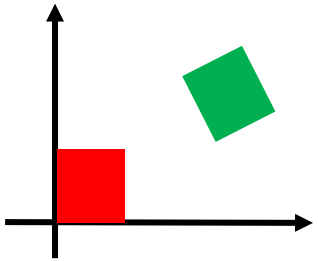


Image formation and representation

2D Transformations

Euclidean $P_t = R \cdot P_s + \vec{t}$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x_s \\ y_s \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Similarity $P_t = s\,R \cdot P_s + \vec{t}$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = s \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x_s \\ y_s \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} sR & \vec{t} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$

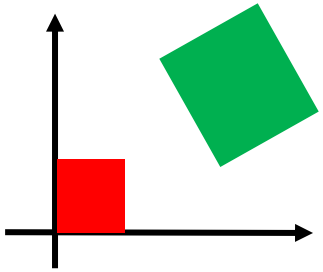


Image formation and representation

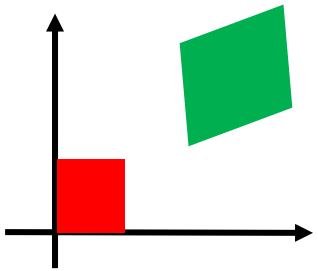
2D Transformations

Affine

$$P_t = A \cdot P_s + \vec{t}$$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix} \begin{pmatrix} x_s \\ y_s \end{pmatrix} + \begin{pmatrix} a_{02} \\ a_{12} \end{pmatrix}$$

$$\begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Projective
or
Homography

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} \dfrac{h_{00}x_s + h_{01}y_s + h_{02}}{h_{20}x_s + h_{21}y_s + h_{22}} \\ \\ \dfrac{h_{00}x_s + h_{01}y_s + h_{02}}{h_{20}x_s + h_{21}y_s + h_{22}} \end{pmatrix}$$

$$\begin{bmatrix} x_t \\ y_t \\ w \end{bmatrix} = \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$

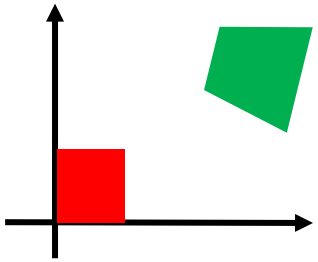


Image formation and representation

2D Transformations

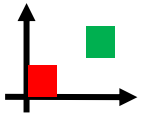
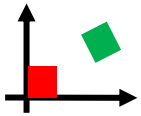
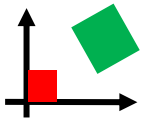
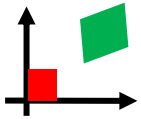
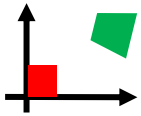
Transformation	Matrix	#DoF	
Translation	$\begin{bmatrix} I_{2\times2} & \vec{t}_{2x1} \\ 0 & 1 \end{bmatrix}$	**2**	
Euclidean	$\begin{bmatrix} R_{2\times2} & \vec{t}_{2x1} \\ 0 & 1 \end{bmatrix}$	**3**	
Similarity	$\begin{bmatrix} sR_{2\times2} & \vec{t}_{2x1} \\ 0 & 1 \end{bmatrix}$	**4**	
Affine	$\begin{bmatrix} A_{2\times2} & \vec{t}_{2x1} \\ 0 & 1 \end{bmatrix}$	**6**	
Projective	$[H_{3\times3}]$	**8**	

Image formation and representation

3D Transformations

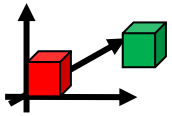
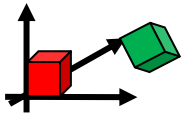
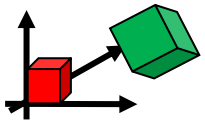
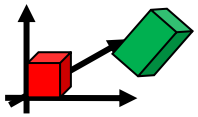
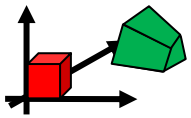
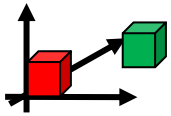
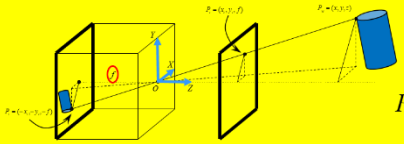
Transformation	Matrix	#DoF	
Translation	$\begin{bmatrix} I_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	**3**	
Euclidean	$\begin{bmatrix} R_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	**6**	
Similarity	$\begin{bmatrix} sR_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	**7**	
Affine	$\begin{bmatrix} A_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	**12**	
Projective	$[H_{4\times4}]$	**15**	

Image formation and representation

3D Transformations

Transformation	Matrix	#DoF
Translation	$\begin{bmatrix} I_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	3
Euclidean	$\begin{bmatrix} R_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	6
Similarity	$\begin{bmatrix} sR_{3\times3} & \vec{t}_{3x} \\ 0 & \end{bmatrix}$	
Affine	$\begin{bmatrix} A_{3\times3} \\ 0 \end{bmatrix}$	
Projective	$[H_{4\times4}]$	





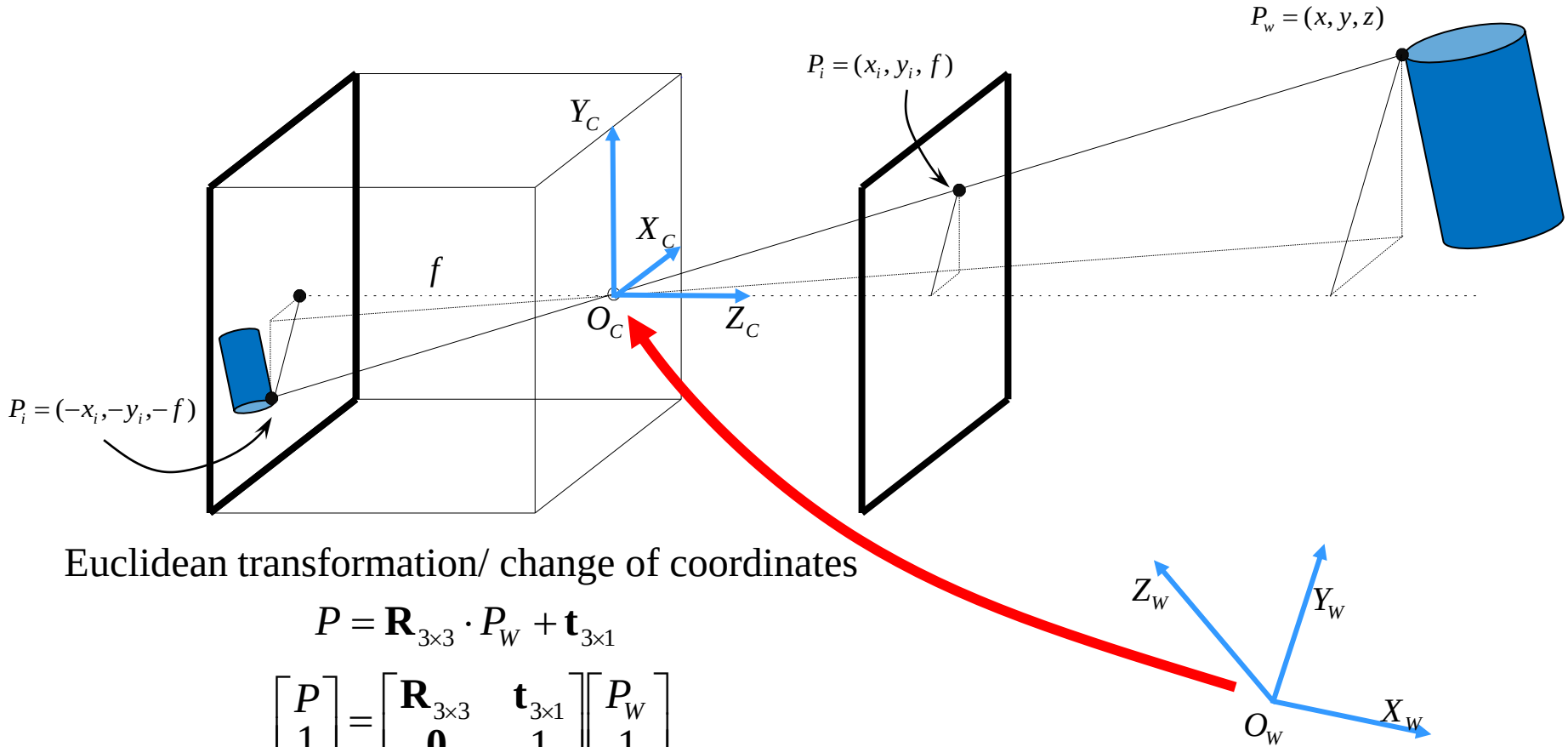
$$P_i^j = \begin{bmatrix} \alpha & 0 & x_0 & 0 \\ 0 & \beta & y_0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = K \cdot P_w^{\,j}$$

- Measure enough pairs $\left(P_i^j ; P_w^{\,j}\right) \quad j = 1..N$
- Obtain enough linear equations in the parameters of K
- Solve for the parameters of K

Problem : measuring P_i^j is easy, but how can we measure $P_w^{\,j}$?

Image formation and representation

Camera *extrinsic* parameters



Euclidean transformation/ change of coordinates

$$P = \mathbf{R}_{3\times3} \cdot P_W + \mathbf{t}_{3\times1}$$

$$\begin{bmatrix} P \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{R}_{3\times3} & \mathbf{t}_{3\times1} \\ \mathbf{0} & 1 \end{bmatrix}}_{\mathbf{F}} \begin{bmatrix} P_W \\ 1 \end{bmatrix}$$

Image formation and representation

The perspective camera projection matrix

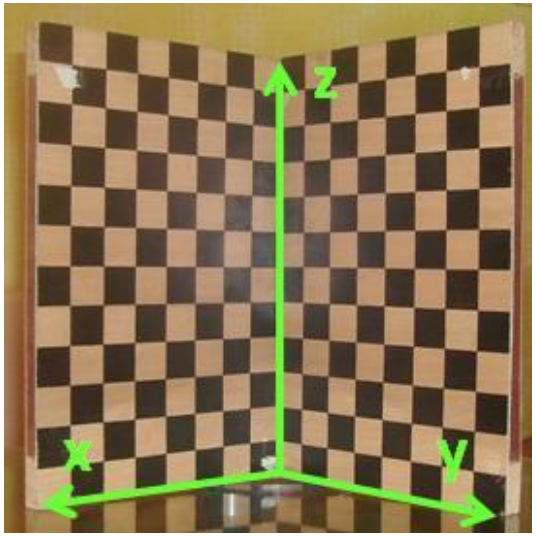
$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{M} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Image formation and representation

Camera calibration

- Measure enough pairs $\left(P_i^{\,j}; P_w^{\,j}\right) \quad j = 1..N \geq 6$
- Estimate **M**
- If needed, decompose M=RQ to estimate the intrinsic and extrinsic parameters separately

$$P_i^{\,j} = \mathbf{M} \cdot P_w^{\,j}$$



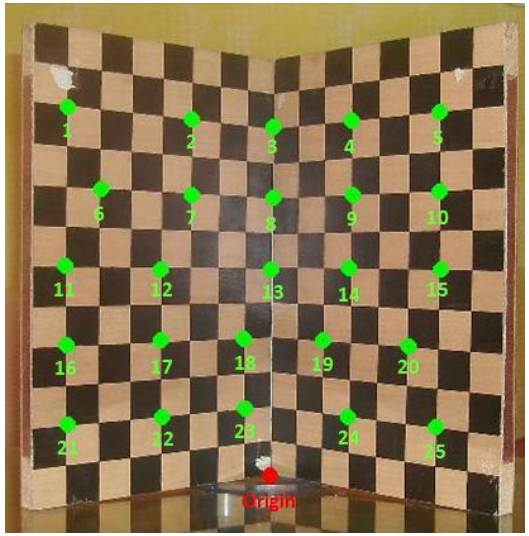


Image formation and representation

The main problem of vision – recovery of structure is ill defined

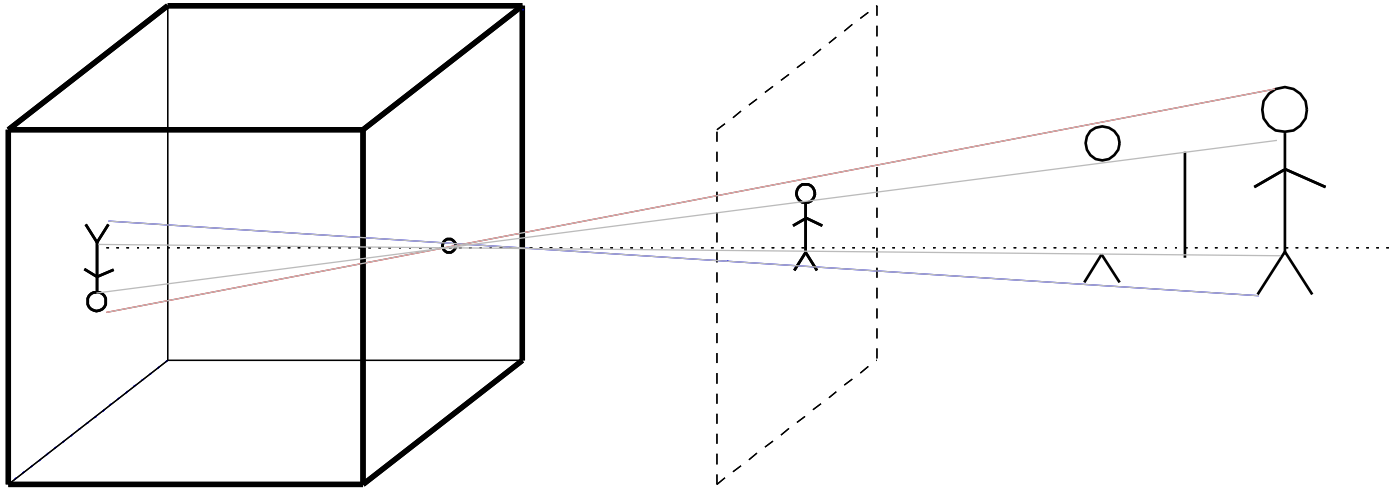


Image formation and representation

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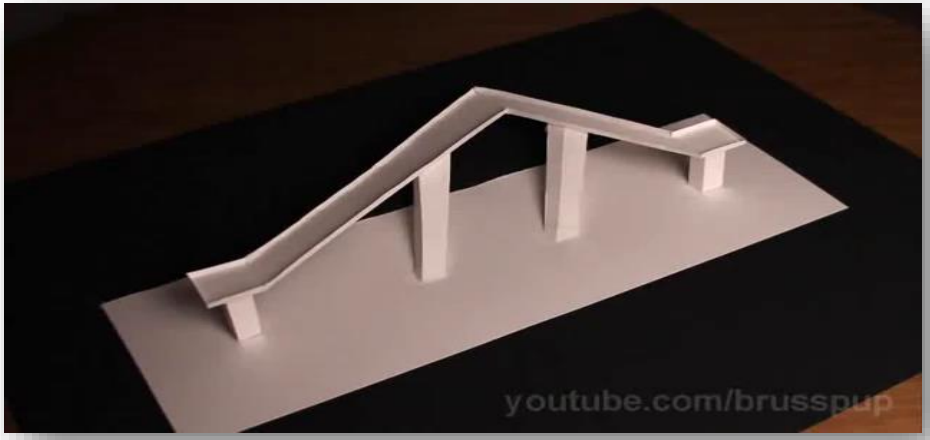

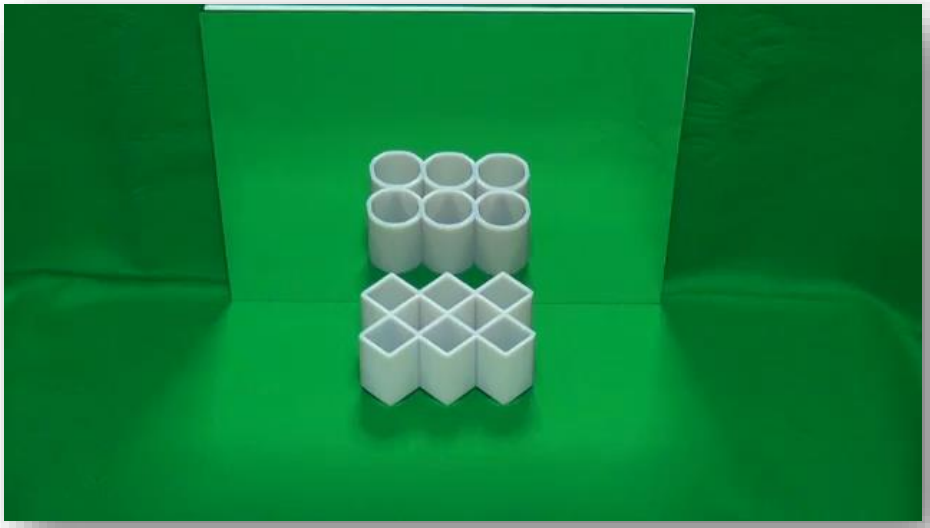

Image formation and representation

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